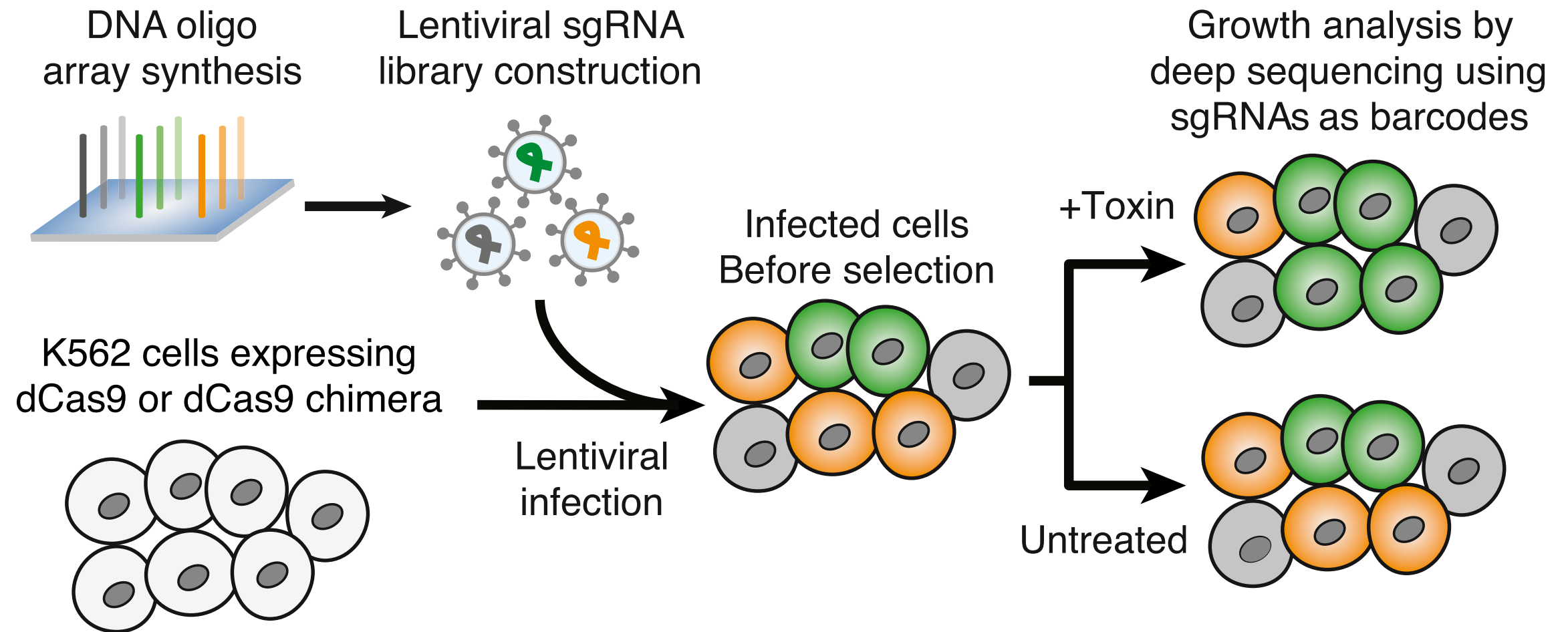


Outline

- Brief review of pooled screening
- Download and install software
- Step 1: counting sgRNAs in raw sequencing files
- Step 2: calculating sgRNA-level and gene-level phenotypes and p-values
- Interpreting key graphs
- Step 3: making custom graphs
- Downstream analysis with GO and StringDB

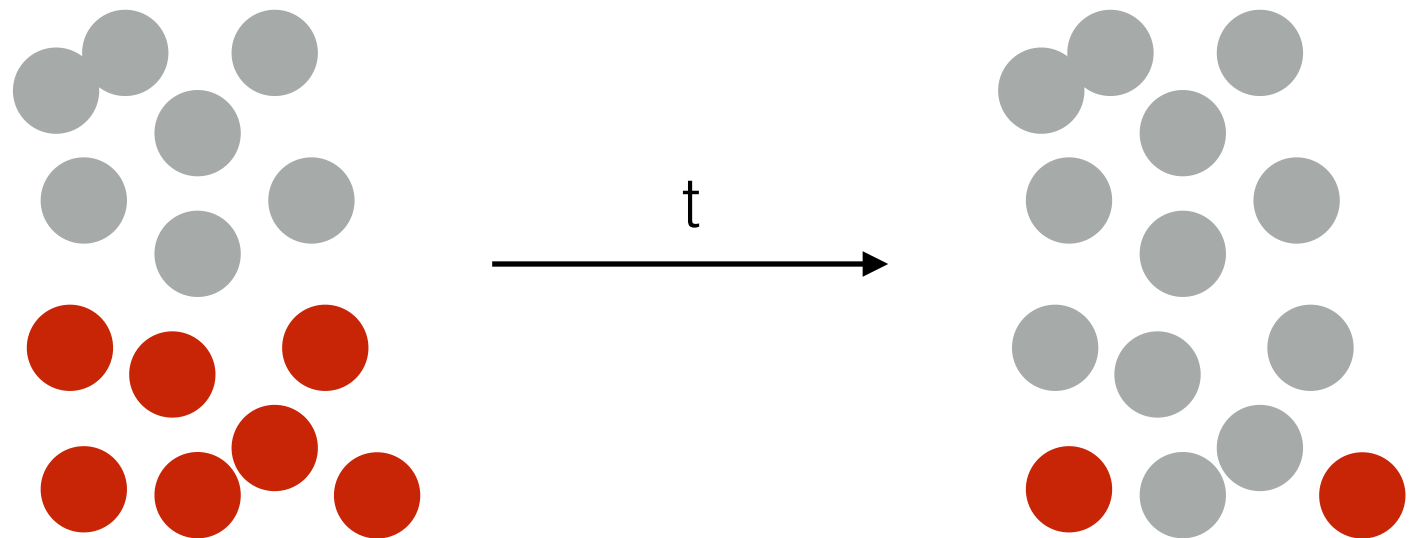


Common types of screens

- Growth based

cells containing:
non-targeting sgRNA

sgRNA targeting
an essential gene

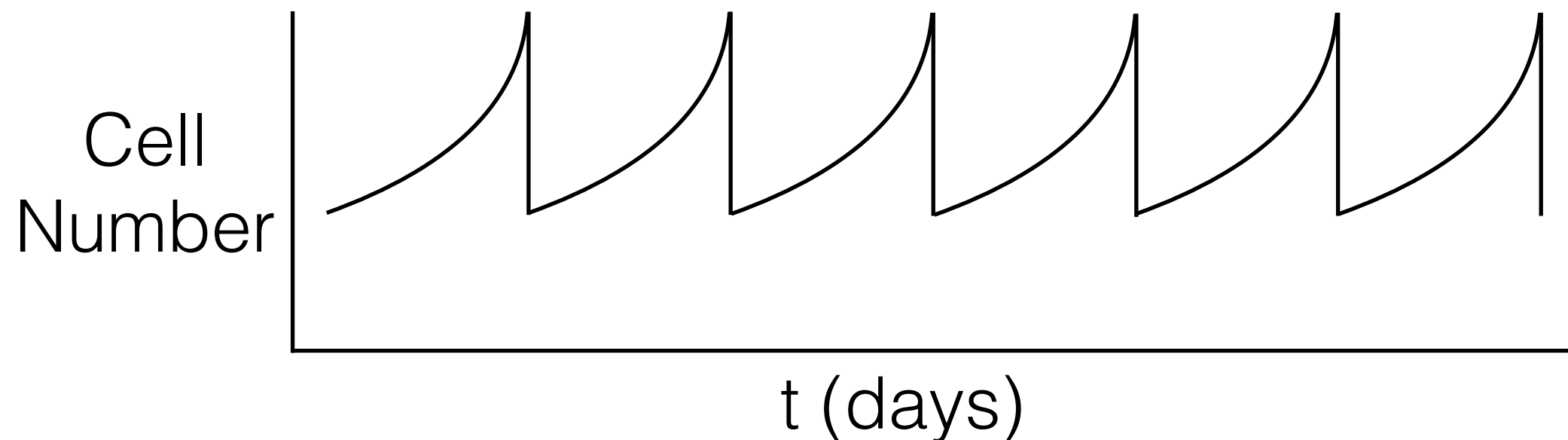
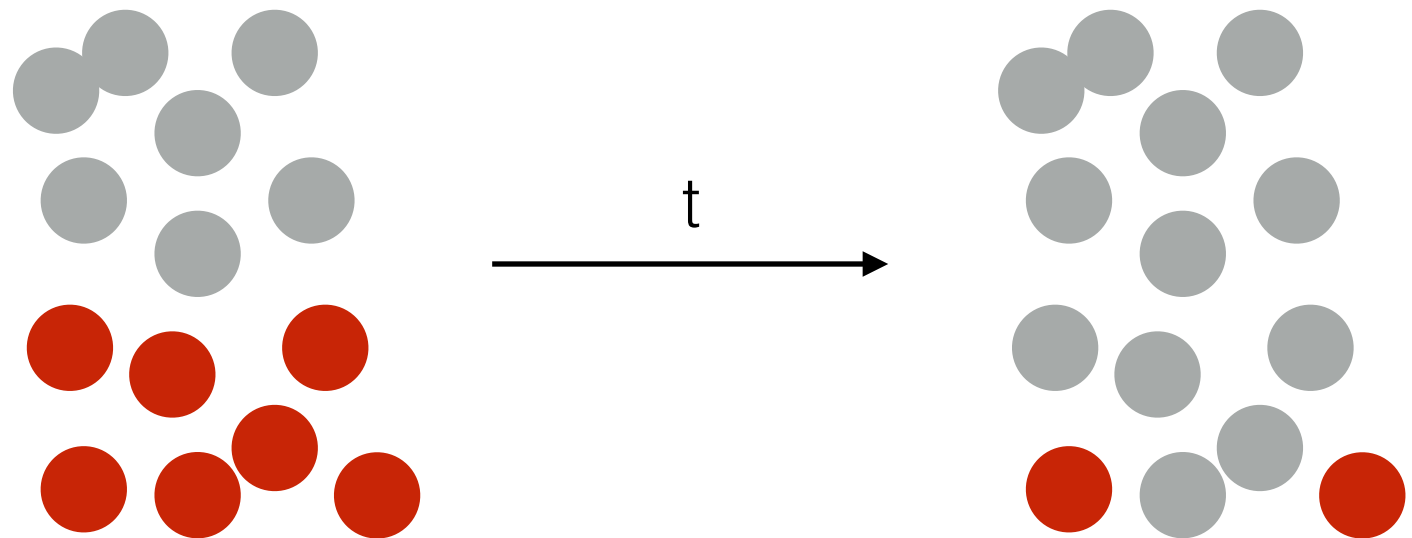


Common types of screens

- Growth based

cells containing:
non-targeting sgRNA

sgRNA targeting
an essential gene

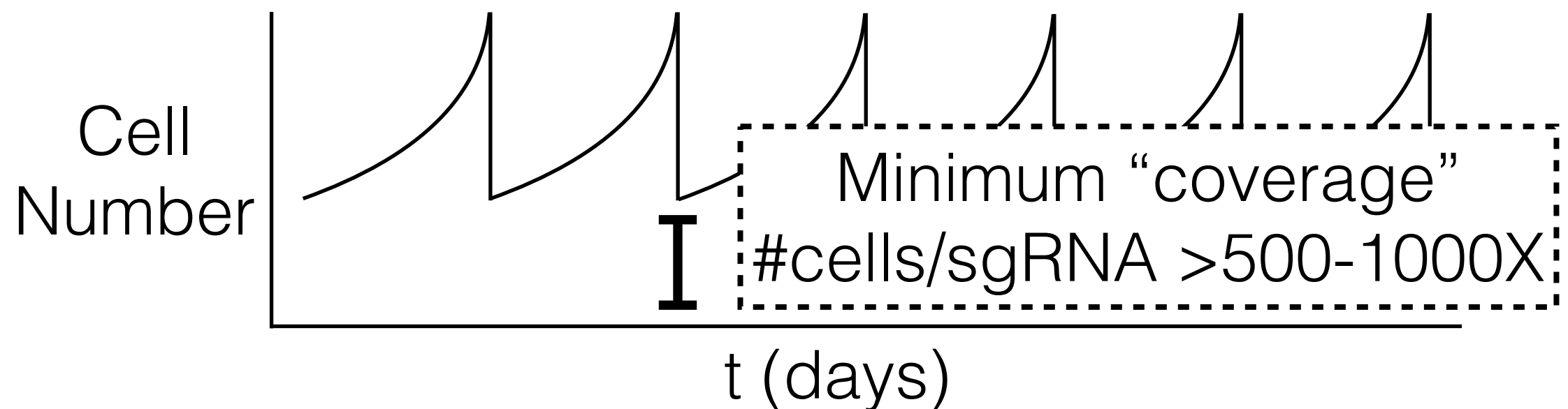
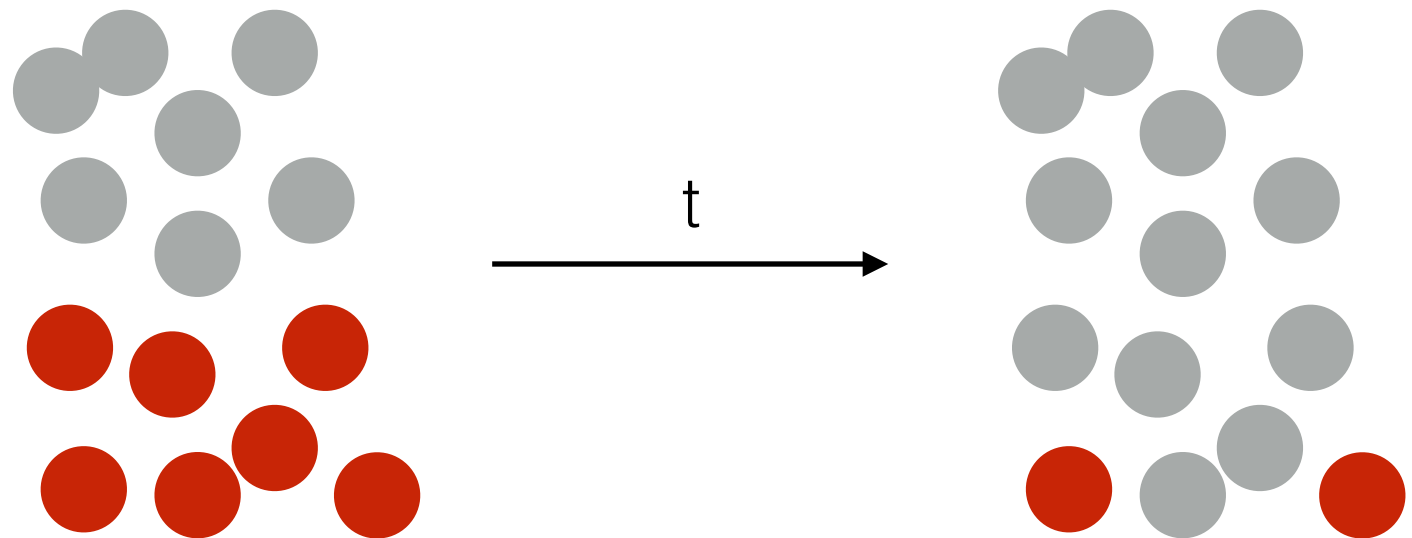


Common types of screens

- Growth based

cells containing:
non-targeting sgRNA

sgRNA targeting
an essential gene

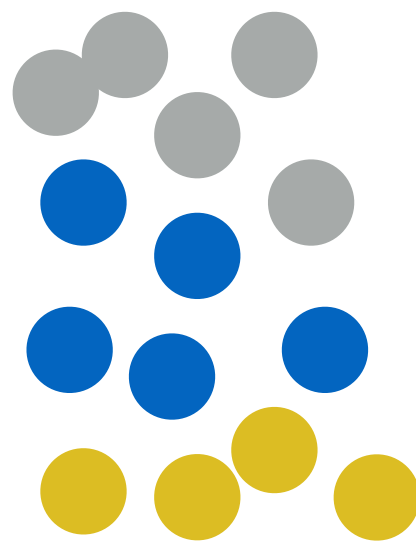


Common types of screens

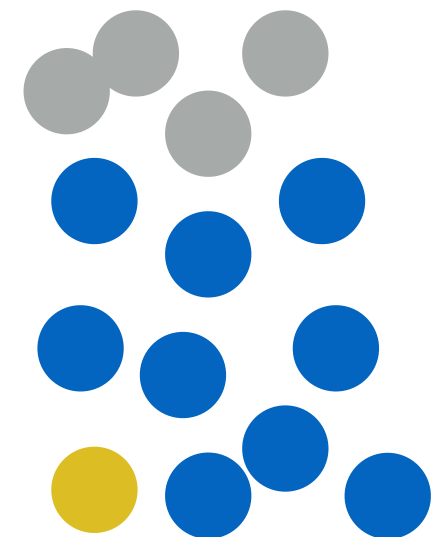
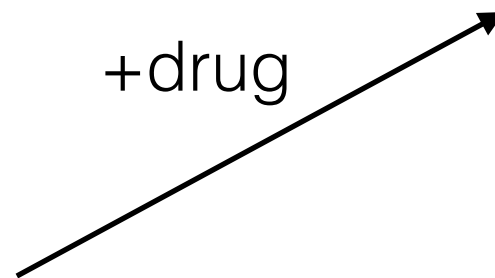
- Growth based
- Drug/Toxin selection

cells containing:
non-targeting sgRNA
sgRNA that promotes
resistance

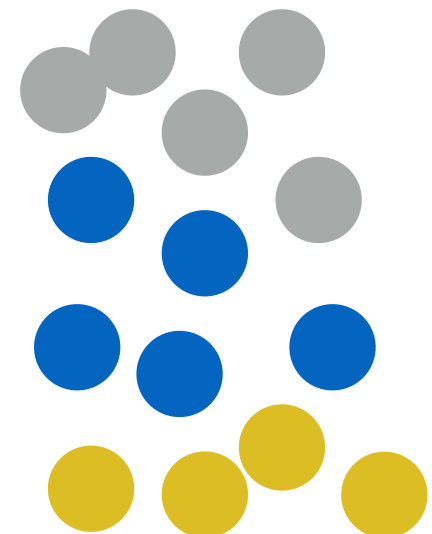
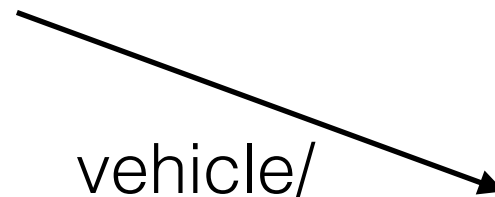
sensitivity



+drug



vehicle/
no drug

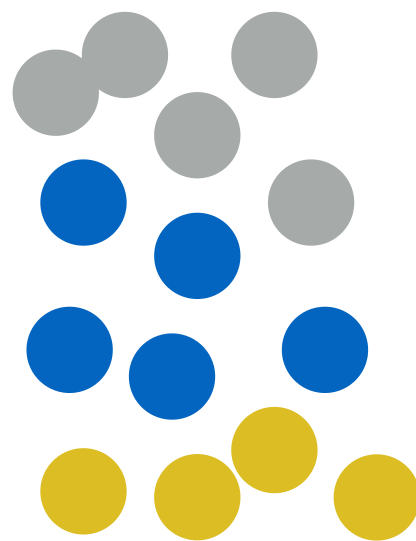


Common types of screens

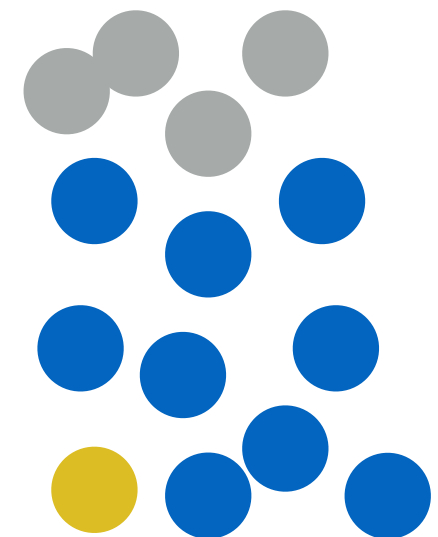
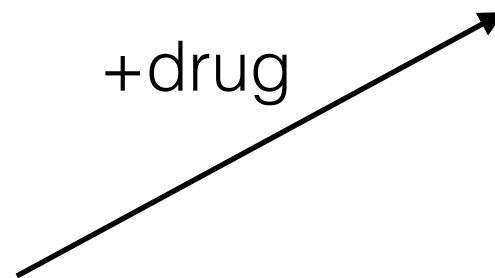
- Growth based
- Drug/Toxin selection

cells containing:
non-targeting sgRNA
sgRNA that promotes
resistance

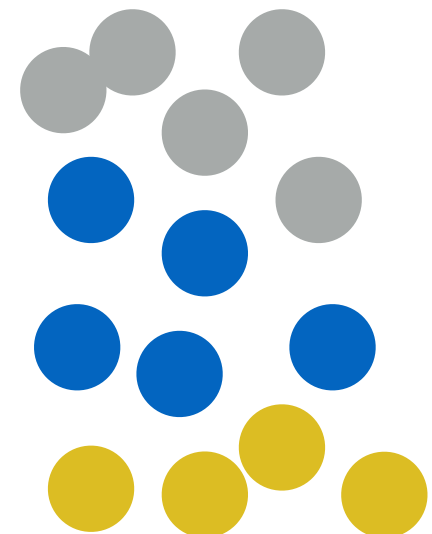
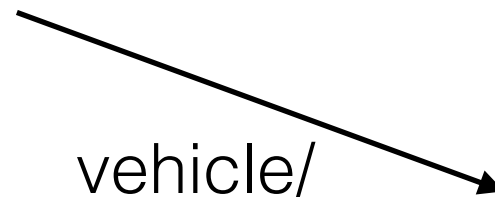
sensitivity



+drug



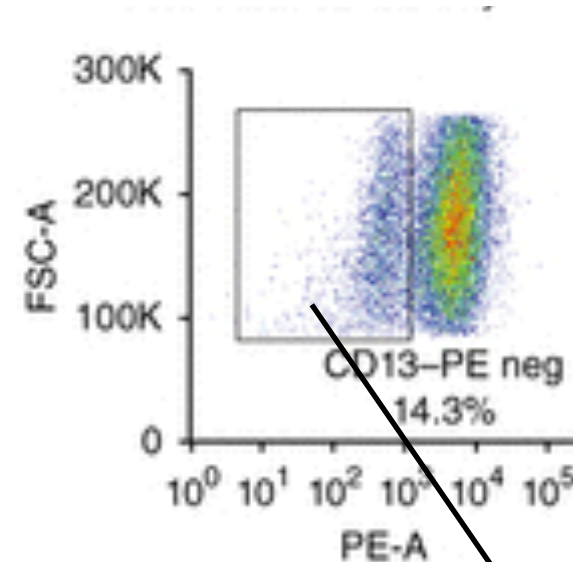
vehicle/
no drug



Minimum coverage is a key consideration
Dosing: LD50? LD90?

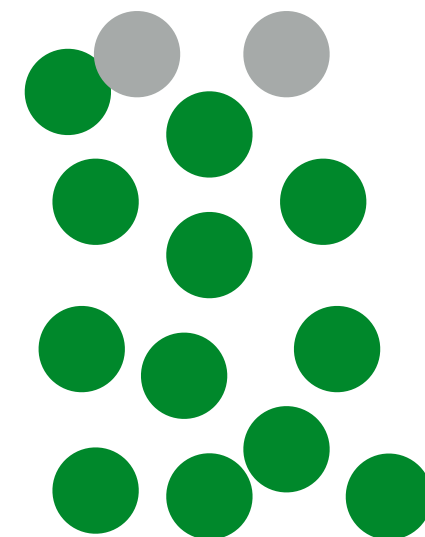
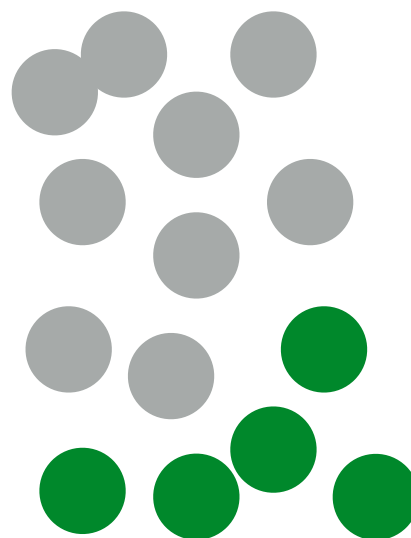
Common types of screens

- Growth based
- Drug/Toxin selection
- FACS sorting



cells containing:
non-targeting sgRNA
sgRNAs targeting CD13

Presorted



How many cells do you
need to grow for a screen?

How many cells do you need to grow for a screen?

```
#genes  
#sgRNAs / gene  
#cells / sgRNA
```

```
multiple conditions  
replicates
```

How many cells do you need to grow for a screen?

#genes
#sgRNAs/gene
#cells/sgRNA

20,000

x10

x1000

200M cells

at minimum

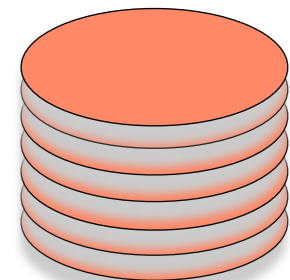
multiple conditions
replicates

How many cells do you need to grow for a screen?

#genes
#sgRNAs/gene
#cells/sgRNA

20,000
x10
x1000
200M cells
at minimum

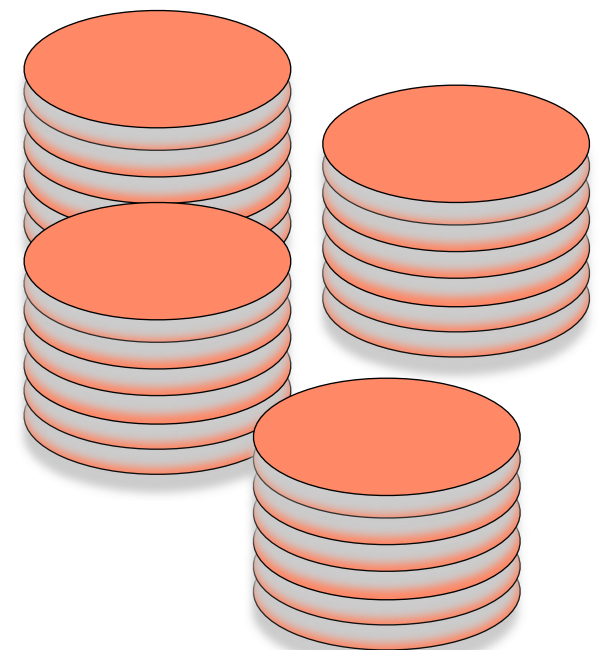
multiple conditions
replicates

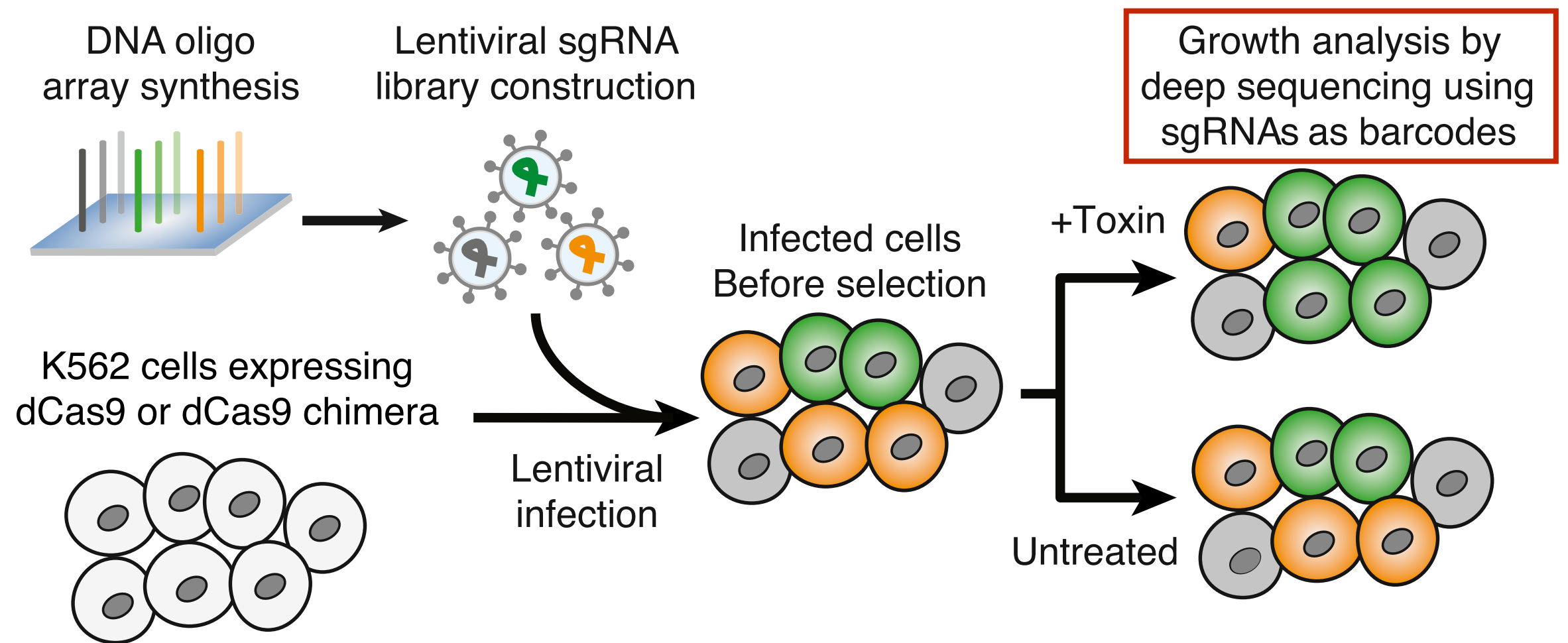


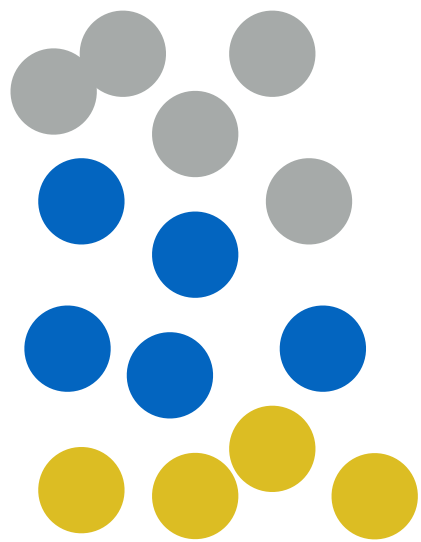
How many cells do you need to grow for a screen?

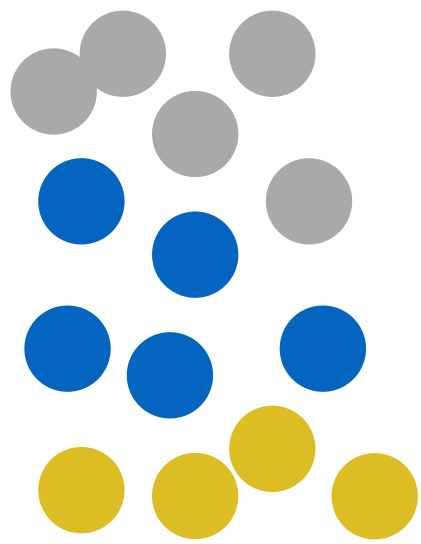
#genes	20,000
#sgRNAs/gene	x10
#cells/sgRNA	x1000
	200M cells
	at minimum

multiple conditions
replicates

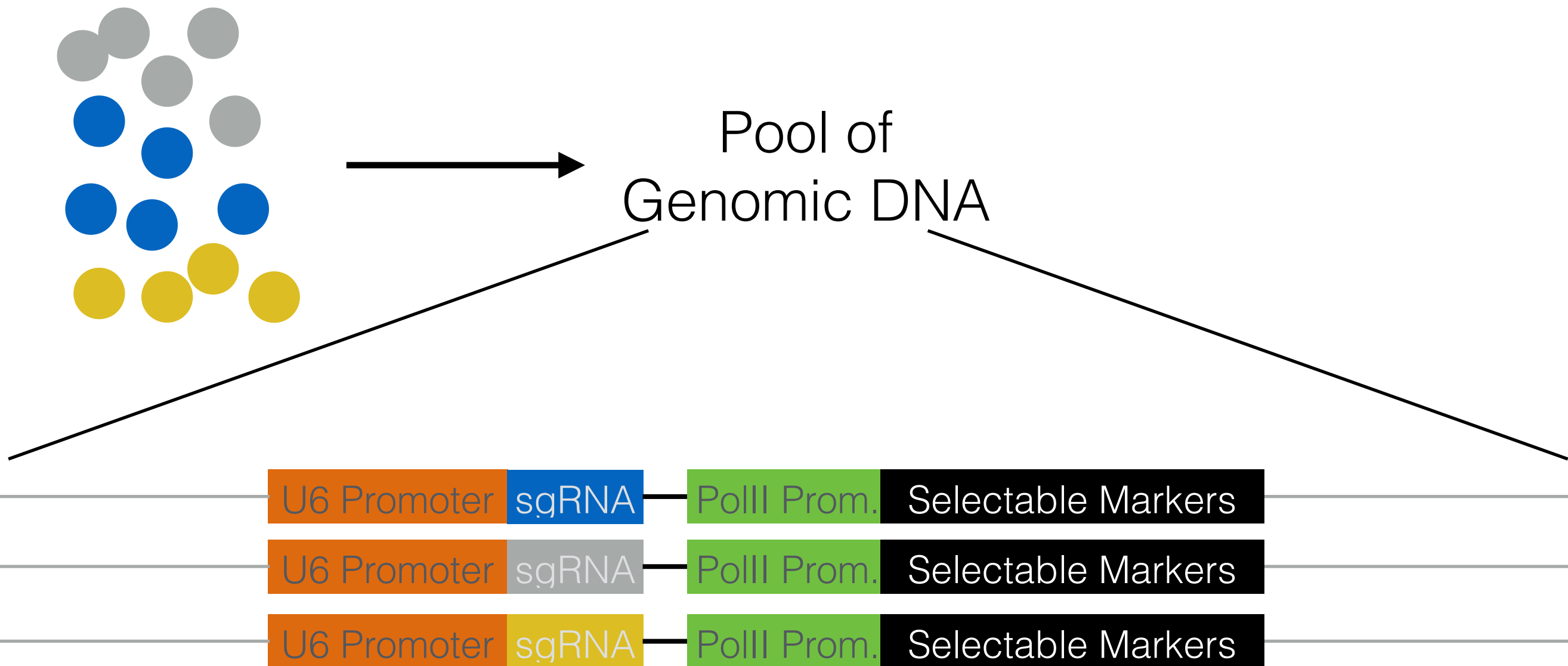


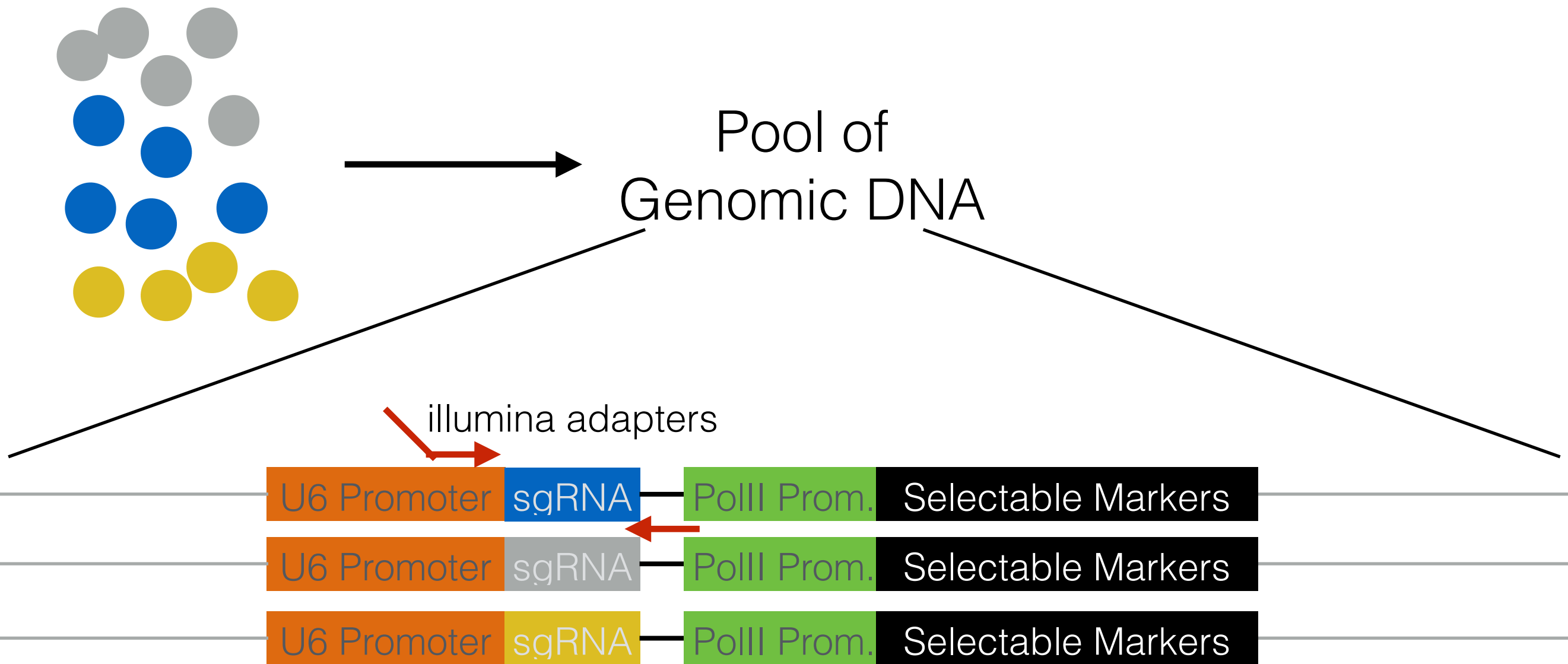


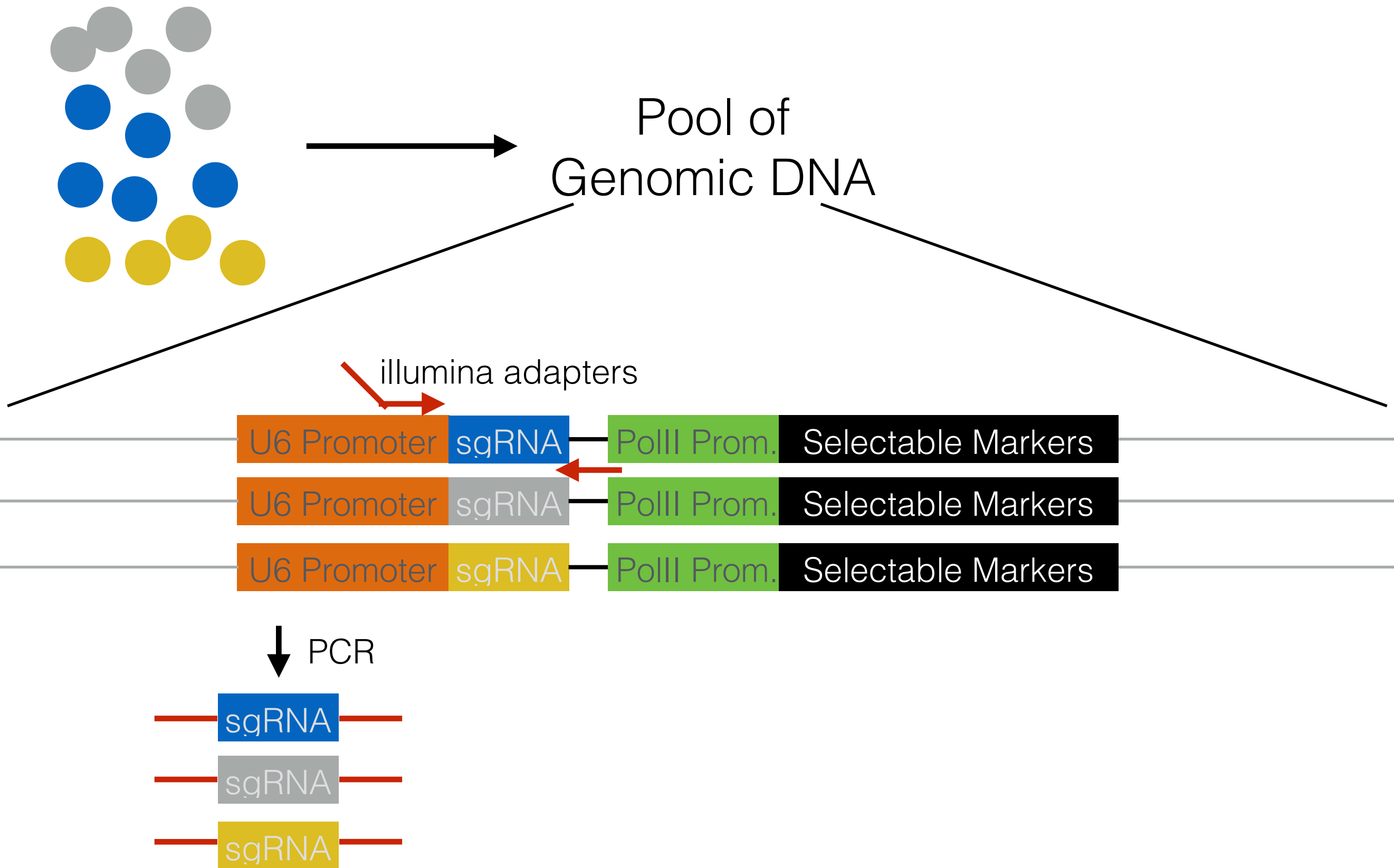


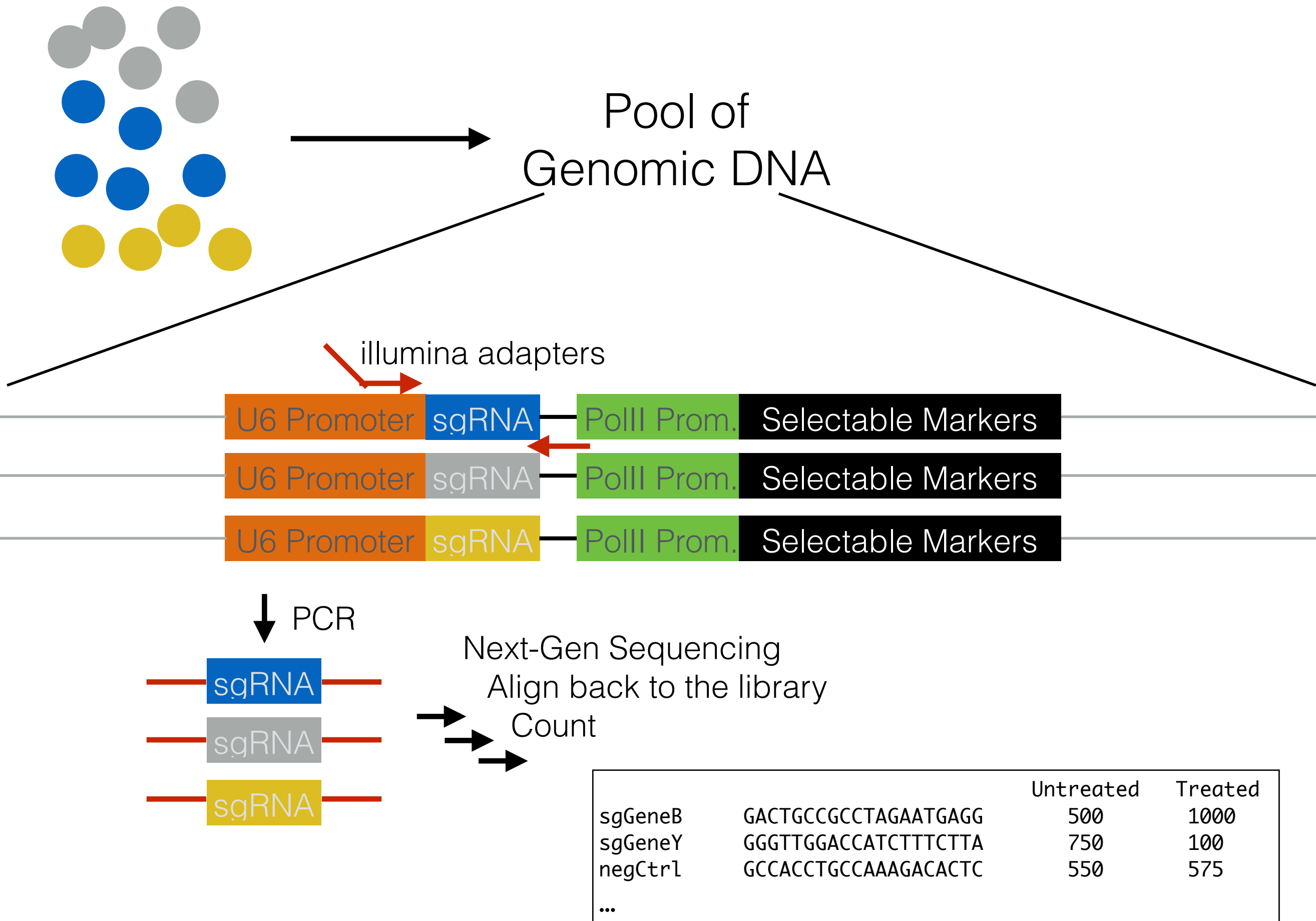


Pool of
Genomic DNA









Setup!

- Install python 2.7, along with ipython, numpy, scipy, pandas, matplotlib packages
- Alternately, download and install Canopy
- <https://store.enthought.com/downloads/>



Enthought Canopy: Easy Python Deployment Plus Integrated Analysis Environment for Scientific Computing, Data Analysis and Engineering

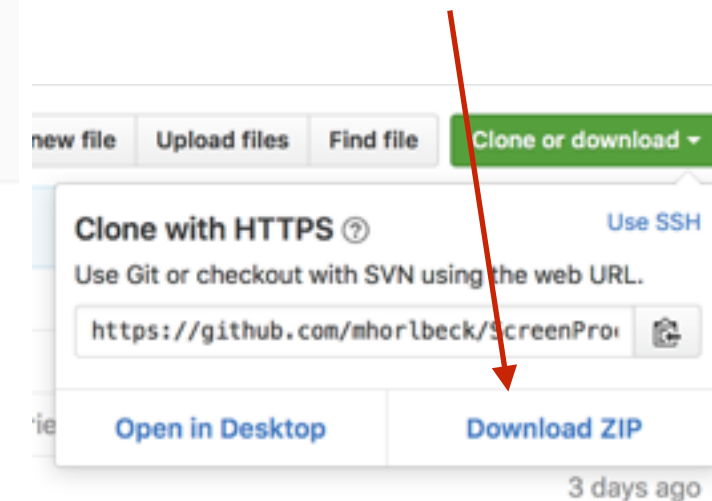
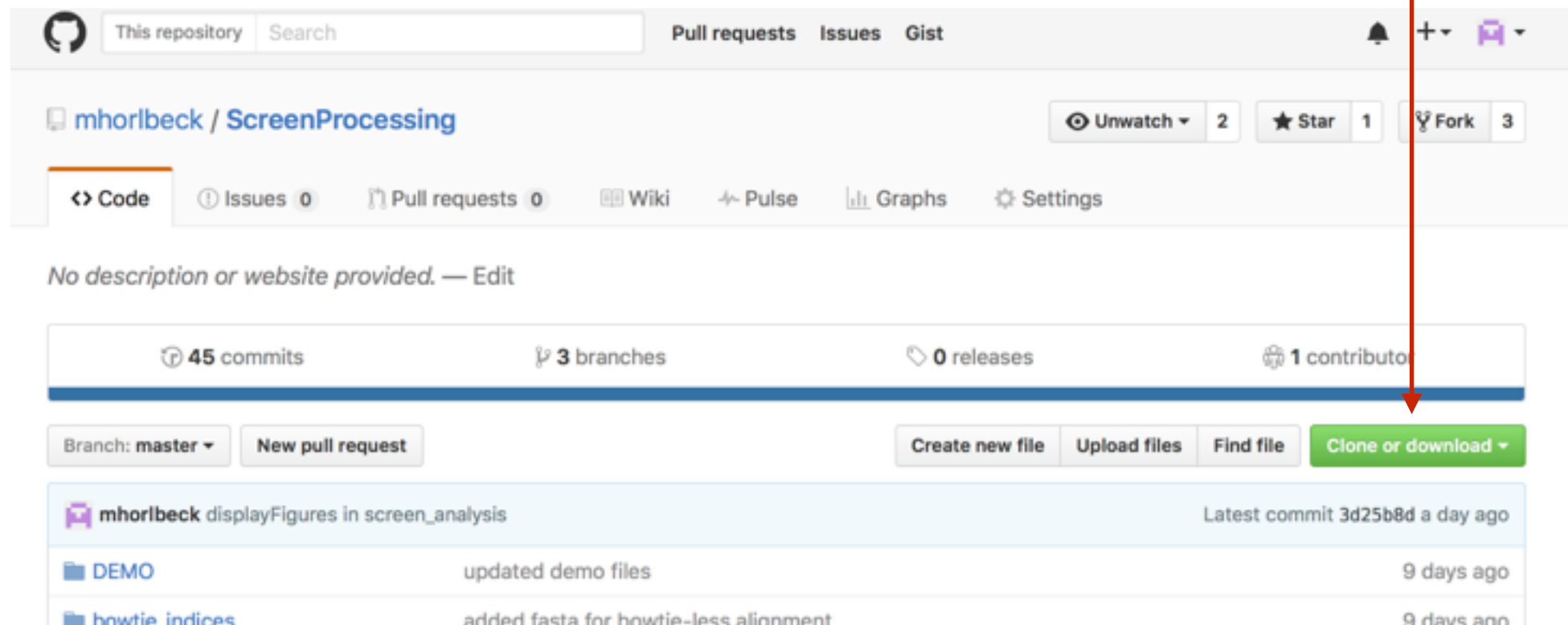
FREE for all users: Canopy Express, which includes access to **100+ of Canopy's most popular Python packages** for scientific computing, data analysis, and engineering PLUS Canopy's integrated analysis environment. Get started today with easy deployment of pre-built, tested, and dependency-aware packages such as **NumPy, SciPy, Pandas, Matplotlib, IPython** and more.

DOWNLOAD
Canopy

By downloading Canopy you acknowledge your acceptance of all the terms and conditions of the applicable license.

Setup!

- Download analysis pipeline scripts
- <https://github.com/mhorlbeck/ScreenProcessing>
- download as zip and unzip on your computer, keeping track of the folder where you unzipped the files



File paths

- You always have a current working directory

- Identify yours with **pwd**

```
[In [1]: pwd  
Out[1]: u'/home/mhorlbeck'
```

- You can change the working directory with **cd**

```
[In [2]: cd home_tmp/mhorlbeck/ScreenProcessing/  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing
```

```
[In [3]: pwd  
Out[3]: u'/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing'
```

- There are “absolute paths” starting with “/” or “C://” and “relative paths” relative to your working directory
- There are several additional shortcuts you can use:
 - . = current directory
 - .. = parent directory
 - ~ = home directory

Running scripts

- Start by **cd**-ing into the unzipped ScreenProcessing directory

```
[In [2]: cd home_tmp/mhorlbeck/ScreenProcessing/  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing  
  
[In [3]: pwd  
Out[3]: u'/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing'
```

- Run scripts using the **run** command—try running the first script we'll use with

run fastqgz_to_counts.py -h

```
[In [5]: run fastqgz_to_counts.py -h  
usage: fastqgz_to_counts.py [-h] [-p PROCESSORS] [--trim_start TRIM_START]  
                           [--trim_end TRIM_END] [--test]  
                           Library_Fasta Out_File_Path Seq_File_Names  
                           [Seq_File_Names ...]  
  
Process raw sequencing data from screens to counts files in parallel  
  
positional arguments:  
  Library_Fasta      Fasta file of expected library reads.  
  Out_File_Path      Directory where output files should be written.  
  Seq_File_Names     Name(s) of sequencing file(s). Unix wildcards can be  
                    used to select multiple files at once. The script will  
                    search for all *.fastq.gz, *.fastq, and  
                    *.fa(/fasta/fna) files with the given wildcard name.  
  
optional arguments:  
  -h, --help          show this help message and exit  
  -p PROCESSORS, --processors PROCESSORS  
  --trim_start TRIM_START  
  --trim_end TRIM_END  
  --test              Run the entire script on only the first 10000 reads of  
                    each file. Be sure to delete or move all test files  
                    before re-running script as they will not be  
                    overwritten.
```

Step 1: fastqgz_to_counts.py

counting sgRNAs in raw sequencing files



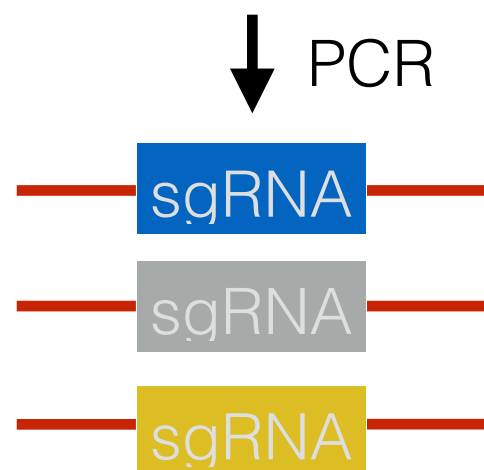
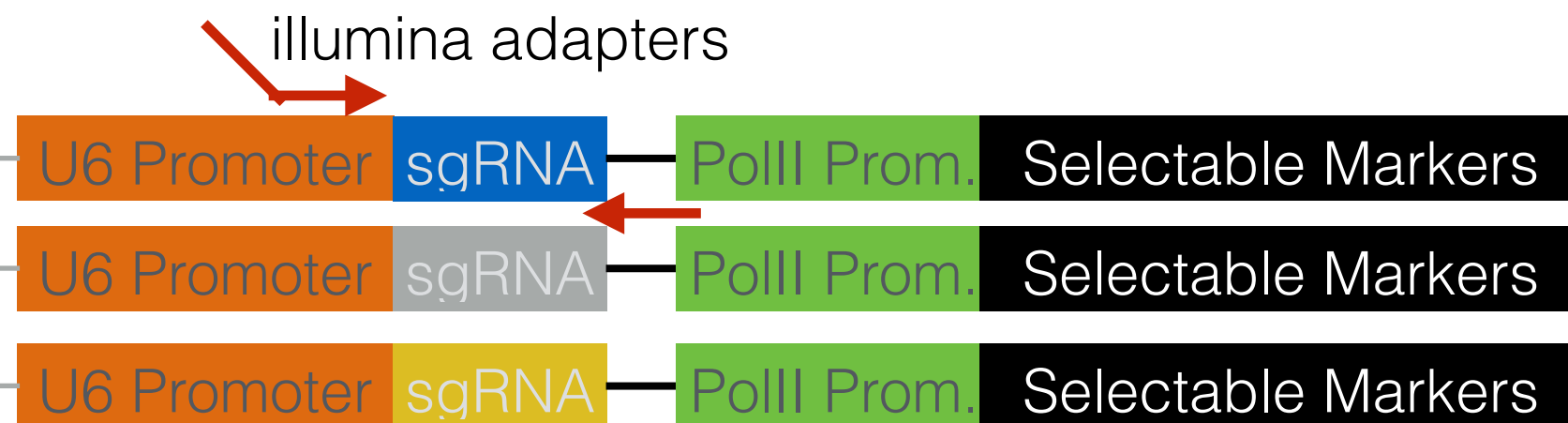
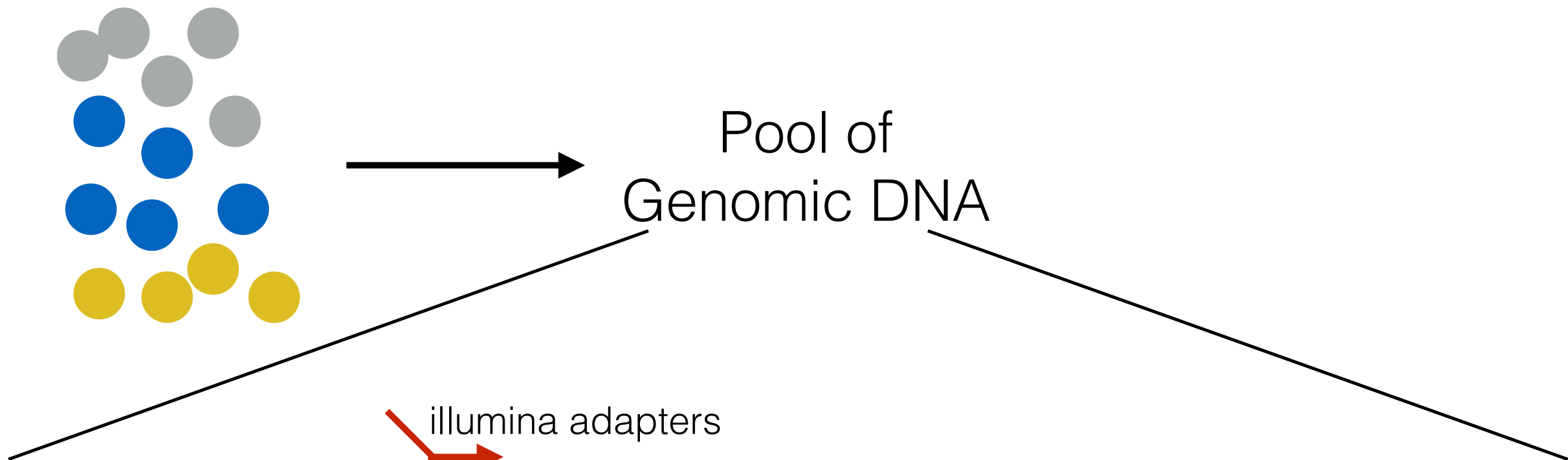
Step 2: process_experiments.py

calculating sgRNA-level and gene-level phenotypes and p-values



Step 3: screen_analysis.py

making custom graphs



Next-Gen Sequencing
Align back to the library
Count

		Untreated	Treated
sgGeneB	GACTGCCGCCTAGAAATGAGG	500	1000
sgGeneY	GGGTTGGACCATCTTTCTTA	750	100
negCtrl	GCCACCTGCCAAAGACACTC	550	575
...			

Raw sequencing files

- Let's take a look at the data we get back from the sequencer—look in Demo/Step1/Sequencing_files
- open any file named _____.fastq in a text editor

Raw sequencing files

- Let's take a look at the data we get back from the sequencer—look in Demo/Step1/Sequencing_files
 - open any file named _____.fastq in a text editor

```
@HWI-ST640:851:C4U9HACXX:8:1101:1149:2177 1:N:0:ATCACG
GGGACCCAGCCTAGACGTGTGTTTAAGAGCTAAGCTGGAAACAGCATAGCA
+
@@@DBDDD3CFB?F9:C<C+ACGGII9C:EGGIDFGIGFGFFIBGFIIIII
@HWI-ST640:851:C4U9HACXX:8:1101:1160:2199 1:N:0:ATCACG
GTTCCAGCCCCACCGCACCGCGTTTAAGAGCTAAGCTGGAAACAGCATAGC
+
@@@DDFFFHHHGHGIIJJJJIJGHHIJJ9BGHGFGIJJHBCHHHI@GGIGCE
@HWI-ST640:851:C4U9HACXX:8:1101:1225:2207 1:N:0:ATCACG
GCCTGTGACTTTAAGGGGCAGTTTAAGAGCTAAGCTGGAAACAGCATAGCA
+
•RRFFFFHHHHGCT1111T11TT1T1AH?GHHTDHHGGFHH1TGH1T1111
```

Raw sequencing files

- Let's take a look at the data we get back from the sequencer—look in Demo/Step1/Sequencing_files
 - open any file named _____.fastq in a text editor

```
@HWI-ST640:851:C4U9HACXX:8:1101:1149:2177 1:N:0:ATCACG
GGGACCCAGCCTAGACGTGTGTTTAAGAGCTAAGCTGGAAACAGCATAGCA
+
@@@DBDDD3CFB?F9:C<C+ACGGII9C:EGGIDFGIGFGFFIBGFIIIII
@HWI-ST640:851:C4U9HACXX:8:1101:1160:2199 1:N:0:ATCACG
GTTCCAGCCCCACCGCACCGCGTTTAAGAGCTAAGCTGGAAACAGCATAGC
+
@@@DDFFFHHHGHGIIJJJJIJGHHIJJ9BGHGFGIJJHBCHHHI@GGIGCE
@HWI-ST640:851:C4U9HACXX:8:1101:1225:2207 1:N:0:ATCACG
GCCTGTGACTTTAAGGGGCAGTTTAAGAGCTAAGCTGGAAACAGCATAGCA
+
•RRFFFFFFHHHHGCT1111T11TT1T1AH?GHHTDHHGGFHH1TGH1T1111
```

Raw sequencing files

- Let's take a look at the data we get back from the sequencer—look in Demo/Step1/Sequencing_files
 - open any file named _____.fastq in a text editor

```
@HWI-ST640:851:C4U9HACXX:8:1101:1149:2177 1:N:0:ATCACG
GGGACCCAGCCTAGACGTGTGTTTAAGAGCTAAGCTGGAAACAGCATAGCA
+
@@@DBDDD3CFB?F9:C<C+ACGGII9C:EGGIDFGIGFGFFIBGFIIIII
@HWI-ST640:851:C4U9HACXX:8:1101:1160:2199 1:N:0:ATCACG
GTTCCAGCCCCACCGCACCGCGTTTAAGAGCTAAGCTGGAAACAGCATAGC
+
@@@DDFFFHHHGHGIIJJJJIJGHHIJJ9BGHGFGIJJHBCHHHI@GGIGCE
@HWI-ST640:851:C4U9HACXX:8:1101:1225:2207 1:N:0:ATCACG
GCCTGTGACTTTAAGGGGCAGTTTAAGAGCTAAGCTGGAAACAGCATAGCA
+
•RRFFFFFFHHHHGCT1111T11TT1T1AH?GHHTDHHGGEHH1TGH1T1111
```

sequence ID

Raw sequencing files

- Let's take a look at the data we get back from the sequencer—look in Demo/Step1/Sequencing_files
 - open any file named _____.fastq in a text editor

```
@HWI-ST640:851:C4U9HACXX:8:1101:1149:2177 1:N:0:ATCACG
GGGACCCAGCCTAGACGTGTGTTTAAGAGCTAAGCTGGAAACAGCATAGCA
+
@@@DBDDD3CFB?F9:C<C+ACGGII9C:EGGIDFGIGFGFFIBGFIIIII
@HWI-ST640:851:C4U9HACXX:8:1101:1160:2199 1:N:0:ATCACG
GTTCCAGCCCCACCGCACCGCGTTTAAGAGCTAAGCTGGAAACAGCATAGC
+
@@@DDFFFHHHGHGIIJJJJIJGHHIJJ9BGHGFGIJJHBCHHHI@GGIGCE
@HWI-ST640:851:C4U9HACXX:8:1101:1225:2207 1:N:0:ATCACG
GCCTGTGACTTTAAGGGGCAGTTTAAGAGCTAAGCTGGAAACAGCATAGCA
+
•RRFFFFFFHHHHGCT1111T11TT1T1AH?GHHTDHHGGEHH1TGH1T1111
```

sequence ID
sequence read

Raw sequencing files

- Let's take a look at the data we get back from the sequencer—look in Demo/Step1/Sequencing_files
 - open any file named _____.fastq in a text editor

```
@HWI-ST640:851:C4U9HACXX:8:1101:1149:2177 1:N:0:ATCACG
GGGACCCAGCCTAGACGTGTGTTTAAGAGCTAAGCTGGAAACAGCATAGCA
+
@@@DBDDD3CFB?F9:C<C+ACGGII9C:EGGIDFGIGFGFFIBGFIIIII
@HWI-ST640:851:C4U9HACXX:8:1101:1160:2199 1:N:0:ATCACG
GTTCCAGCCCCACCGCACCGCGTTTAAGAGCTAAGCTGGAAACAGCATAGC
+
@@@DDFFFHHHGHGIIJJJJIJGHHIJJ9BGHGFGIJJHBCHHHI@GGIGCE
@HWI-ST640:851:C4U9HACXX:8:1101:1225:2207 1:N:0:ATCACG
GCCTGTGACTTTAAGGGGCAGTTTAAGAGCTAAGCTGGAAACAGCATAGCA
+
•RRFFFFFFHHHHGCT1111T11TT1T1AH?GHHTDHHGGEHH1TGH1T1111
```

sequence ID
sequence read

quality scores

Raw sequencing files

- Let's take a look at the data we get back from the sequencer—look in Demo/Step1/Sequencing_files
 - open any file named _____.fastq in a text editor

```
@HWI-ST640:851:C4U9HACXX:8:1101:1149:2177 1:N:0:ATCACG
GGGACCCAGCCTAGACGTGTGTTTAAGAGCTAAGCTGGAAACAGCATAGCA
+
@@@DBDDD3CFB?F9:C<C+ACGGII9C:EGGIDFGIGFGFFIBGFIIIII
@HWI-ST640:851:C4U9HACXX:8:1101:1160:2199 1:N:0:ATCACG
GTTCCAGCCCCACCGCACCGCGTTTAAGAGCTAAGCTGGAAACAGCATAGC
+
@@@DDFFFHHHGHGIIJJJIJGHHIJJ9BGHGFGIJJHBCHHHI@GGIGCE
@HWI-ST640:851:C4U9HACXX:8:1101:1225:2207 1:N:0:ATCACG
GCCTGTGACTTTAAGGGGCAGTTTAAGAGCTAAGCTGGAAACAGCATAGCA
+
•RRFFFFFFHHHHGCT1111T11TT1T1AH?GHHTDHHGGEHH1TGH1T1111
```

sequence ID
sequence read

quality scores

Raw sequencing files

- Let's take a look at the data we get back from the sequencer—look in Demo/Step1/Sequencing_files
 - open any file named _____.fastq in a text editor

```
@HWI-ST640:851:C4U9HACXX:8:1101:1149:2177 1:N:0:ATCACG
GGGACCCAGCCTAGACGTGTGTTTAAGAGCTAAGCTGGAAACAGCATAGCA
+
@@@DBDDD3CFB?F9:C<C+ACGGII9C:EGGIDFGIGFGFFIBGFIIIII
@HWI-ST640:851:C4U9HACXX:8:1101:1160:2199 1:N:0:ATCACG
GTTCCAGCCCCACCGCACCGCGTTTAAGAGCTAAGCTGGAAACAGCATAGC
+
@@@DDFFFHHHGHGIIJJJJIJGHHIJJ9BGHGFGIJJHBCHHHI@GGIGCE
@HWI-ST640:851:C4U9HACXX:8:1101:1225:2207 1:N:0:ATCACG
GCCTGTGACTTTAAGGGGCAGTTTAAGAGCTAAGCTGGAAACAGCATAGCA
+
•RRFFFFFFHHHHGCT1111T11TT1T1AH?GHHTDHHGGEHH1TGH1T1111
```

sequence ID
sequence read

quality scores

library fasta file

- Can be found in /library_reference/CRISPRi_v1_human.trim_1_35.fa

```
>Drug_Targets+Kinase_Phosphatase=CDK10+_89753127.24-all~e39m1  
ATCTGCTCGCACTCCAGATCGTTTAAGAGCTAAG
```

**sgRNA ID
expected sequence**

```
>Drug_Targets+Kinase_Phosphatase=HRH2_-_175108555.26-all~e39m1  
CCTCAGCTTGTTTCTCGCCAGTGTTTAAGAGCTA
```

```
>Apoptosis+Cancer+Other_Cancer=CUL1+_148395795.23-ENST00000602748.1~e39m1  
GGCCGGAACGCACCGCCATGTTTAAGAGCTAAGC
```

```
>Apoptosis+Cancer+Other_Cancer=GTF2H5_-_158589482.24-all~e39m1  
TGGGGTGTGGGAACACTGAGGTTTAAGAGCTAAG
```

```
>Trafficking+Mitochondria+Motility=RHOT1+_30469525.28-all~e39m1  
CCAGCTCCGCCTCCTCCGGCCCAGGTTTAAGAGC
```

```
>Drug_Targets+Kinase_Phosphatase=NDUFC2+_77791131.24-all~e39m1  
GCAATTCTGGTTGGGGAAGTGTTTAAGAGCTAAG
```

```
>Apoptosis+Cancer+Other_Cancer=SREBF1+_17740225.24-all~e39m1  
GACACGAACGCGCGGAGCGGGTTTAAGAGCTAAG
```

Running fastqgz_to_counts

run fastqgz_to_counts.py -h

```
[In [5]: run fastqgz_to_counts.py -h
usage: fastqgz_to_counts.py [-h] [-p PROCESSORS] [--trim_start TRIM_START]
                             [--trim_end TRIM_END] [--test]
                             Library_Fasta Out_File_Path Seq_File_Names
                             [Seq_File_Names ...]

Process raw sequencing data from screens to counts files in parallel

positional arguments:
  Library_Fasta      Fasta file of expected library reads.
  Out_File_Path      Directory where output files should be written.
  Seq_File_Names     Name(s) of sequencing file(s). Unix wildcards can be
                    used to select multiple files at once. The script will
                    search for all *.fastq.gz, *.fastq, and
                    *.fa(/fasta/fna) files with the given wildcard name.

optional arguments:
  -h, --help            show this help message and exit
  -p PROCESSORS, --processors PROCESSORS
                        number of processors to use
  --trim_start TRIM_START
                        start position to trim reads
  --trim_end TRIM_END   end position to trim reads
  --test                Run the entire script on only the first 10000 reads of
                        each file. Be sure to delete or move all test files
                        before re-running script as they will not be
                        overwritten.
```

run fastqgz_to_counts.py -p 6

--trim_start 1 --trim_end 35

library_reference/CRISPRi_v1_human.trim_1_35.fa

Demo/Step1/output

Demo/Step1/Sequencing_files/Demo*

Running fastqgz_to_counts

**run fastqgz_to_counts.py -p 6 --trim_start 1 --trim_end 35 library_reference/
CRISPRi_v1_human.trim_1_35.fa Demo/Step1/output Demo/Step1/Sequencing_files/Demo***

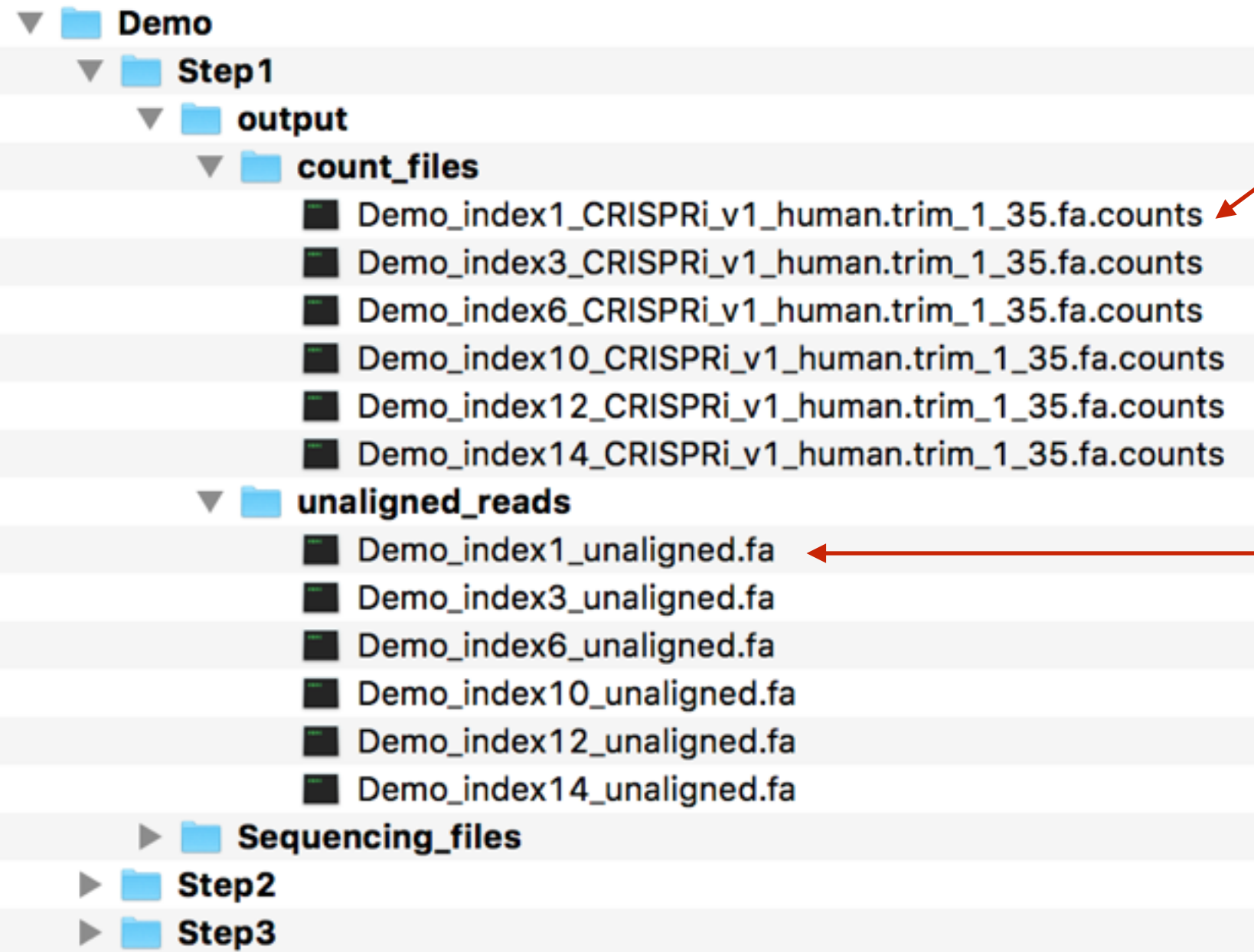
In [7]: run fastqgz_to_counts.py -p 6 --trim_start 1 --trim_end 35 library_reference/CRISPRi_v1_human.trim_1_35.fa Demo/Step1/output
Demo/Step1/Sequencing_files/Demo*

Running fastqgz_to_counts

**run fastqgz_to_counts.py -p 6 --trim_start 1 --trim_end 35 library_reference/
CRISPRi_v1_human.trim_1_35.fa Demo/Step1/output Demo/Step1/Sequencing_files/Demo***

```
In [7]: run fastqgz_to_counts.py -p 6 --trim_start 1 --trim_end 35 library_reference/CRISPRi_v1_human.trim_1_35.fa Demo/Step1/output
Demo/Step1/Sequencing_files/Demo*
[Library file loaded successfully:
  2.09E+05 elements (2.05E+05 unique sequences)  34bp reads expected
Processing Demo/Step1/Sequencing_files/Demo_index1.fastq
Processing Demo/Step1/Sequencing_files/Demo_index10.fastq
Processing Demo/Step1/Sequencing_files/Demo_index12.fastq
Processing Demo/Step1/Sequencing_files/Demo_index14.fastq
Processing Demo/Step1/Sequencing_files/Demo_index3.fastq
Processing Demo/Step1/Sequencing_files/Demo_index6.fastq
Done processing Demo/Step1/Sequencing_files/Demo_index1.fastq
Done processing Demo/Step1/Sequencing_files/Demo_index10.fastq
Done processing Demo/Step1/Sequencing_files/Demo_index6.fastq
Done processing Demo/Step1/Sequencing_files/Demo_index14.fastq
Done processing Demo/Step1/Sequencing_files/Demo_index3.fastq
Done processing Demo/Step1/Sequencing_files/Demo_index12.fastq
Demo/Step1/output/count_files/Demo_index1_CRISPRi_v1_human.trim_1_35.fa.counts:
  2.50E+04 reads  2.05E+04 aligning (81.99%)
Demo/Step1/output/count_files/Demo_index10_CRISPRi_v1_human.trim_1_35.fa.counts:
  2.50E+04 reads  2.04E+04 aligning (81.49%)
Demo/Step1/output/count_files/Demo_index12_CRISPRi_v1_human.trim_1_35.fa.counts:
  2.50E+04 reads  2.03E+04 aligning (81.01%)
Demo/Step1/output/count_files/Demo_index14_CRISPRi_v1_human.trim_1_35.fa.counts:
  2.50E+04 reads  2.03E+04 aligning (81.39%)
Demo/Step1/output/count_files/Demo_index3_CRISPRi_v1_human.trim_1_35.fa.counts:
  2.50E+04 reads  2.03E+04 aligning (81.31%)
Demo/Step1/output/count_files/Demo_index6_CRISPRi_v1_human.trim_1_35.fa.counts:
  2.50E+04 reads  2.03E+04 aligning (81.10%)
Done processing all sequencing files
```

Running fastqgz_to_counts



Apoptosis+Cancer+Other_Cancer=A2M+_9268488.25-all~e39m1
0
Apoptosis+Cancer+Other_Cancer=A2M+_9268495.24-all~e39m1
1
Apoptosis+Cancer+Other_Cancer=A2M+_9268513.26-all~e39m1
0
Apoptosis+Cancer+Other_Cancer=A2M+_9268524.25-all~e39m1
0
Apoptosis+Cancer+Other_Cancer=A2M+_9268659.23-all~e39m1
0
Apoptosis+Cancer+Other_Cancer=A2M+_9268708.26-all~e39m1
0
Apoptosis+Cancer+Other_Cancer=A2M+_9268728.25-all~e39m1
1
Apoptosis+Cancer+Other_Cancer=A2M-_9268620.25-all~e39m1
0
Apoptosis+Cancer+Other_Cancer=A2M-_9268625.25-all~e39m1
0
Apoptosis+Cancer+Other_Cancer=A2M-_9268688.24-all~e39m1
0

>41
ACGTGGAAGCTTACAGCGTTTAAGAGCTAAGC
>77
ATGAGACGGTCTTGACTCGTTTAAGAGCTAAGCT
>85
TCTTGGGCATCAGAGGTGAGAGTTTAAAGCTAA
>109
ACCAGGATGGGCACCAACCGTTTAAGAGCTAAGC
>117
AACTGCCCACGTCCTGGTGTTAAGAGCTAAGCT
>137
AGCGCCACAACGCTCGCGGTTTAAGAGCTAAGCT
>169
AGGACGCAACCAAGCGGTGTTTAAGAGCTAAGC
>173
GATGAGTGAGAGAAAGAAAGTTTAAGAGCTAAGC
>185
CCATGGCTTCGGGAGGCTTAGTTTAAAGCTAAG

Step 1: fastqgz_to_counts.py

counting sgRNAs in raw sequencing files



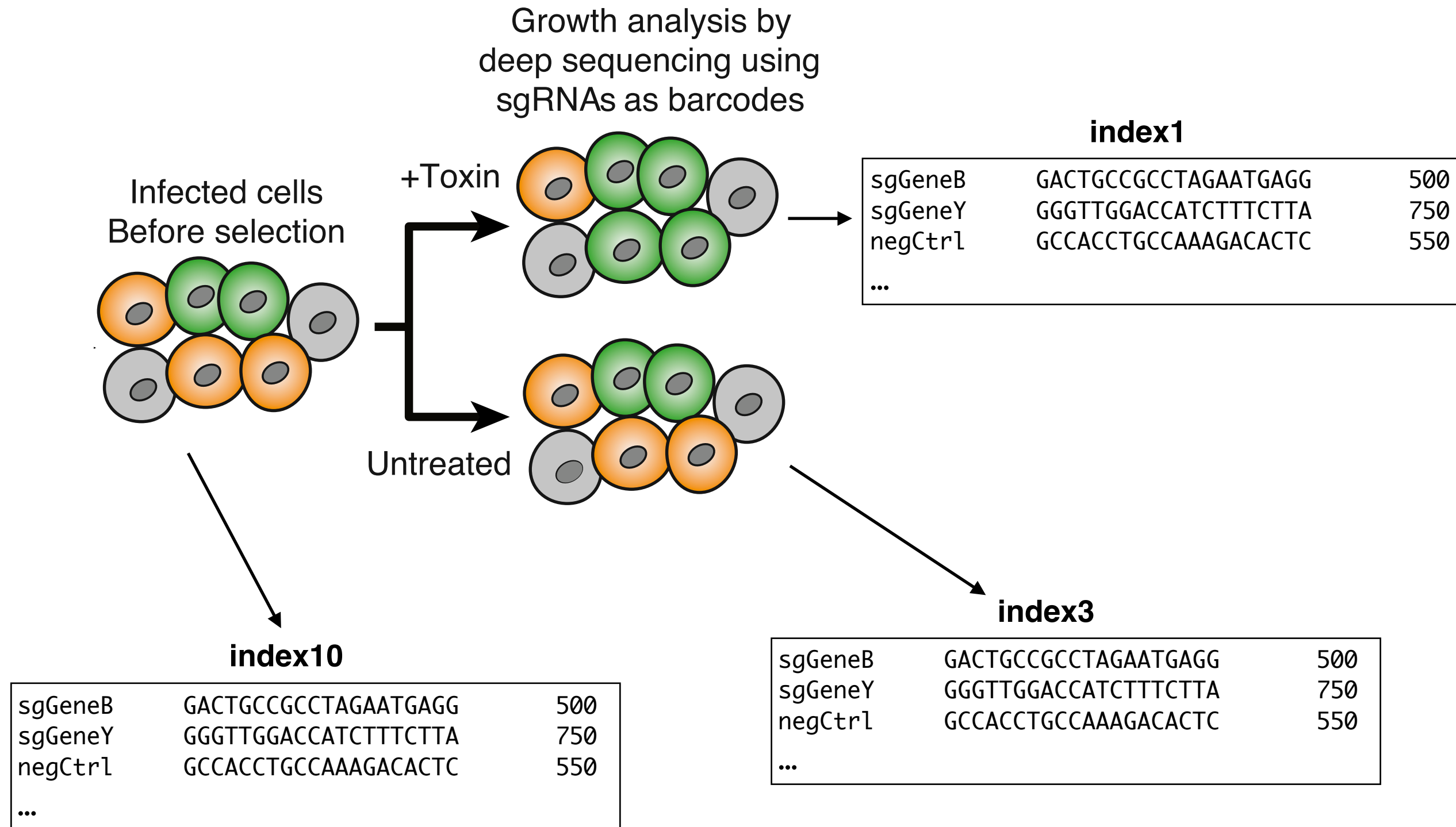
Step 2: process_experiments.py

calculating sgRNA-level and gene-level phenotypes and p-values



Step 3: screen_analysis.py

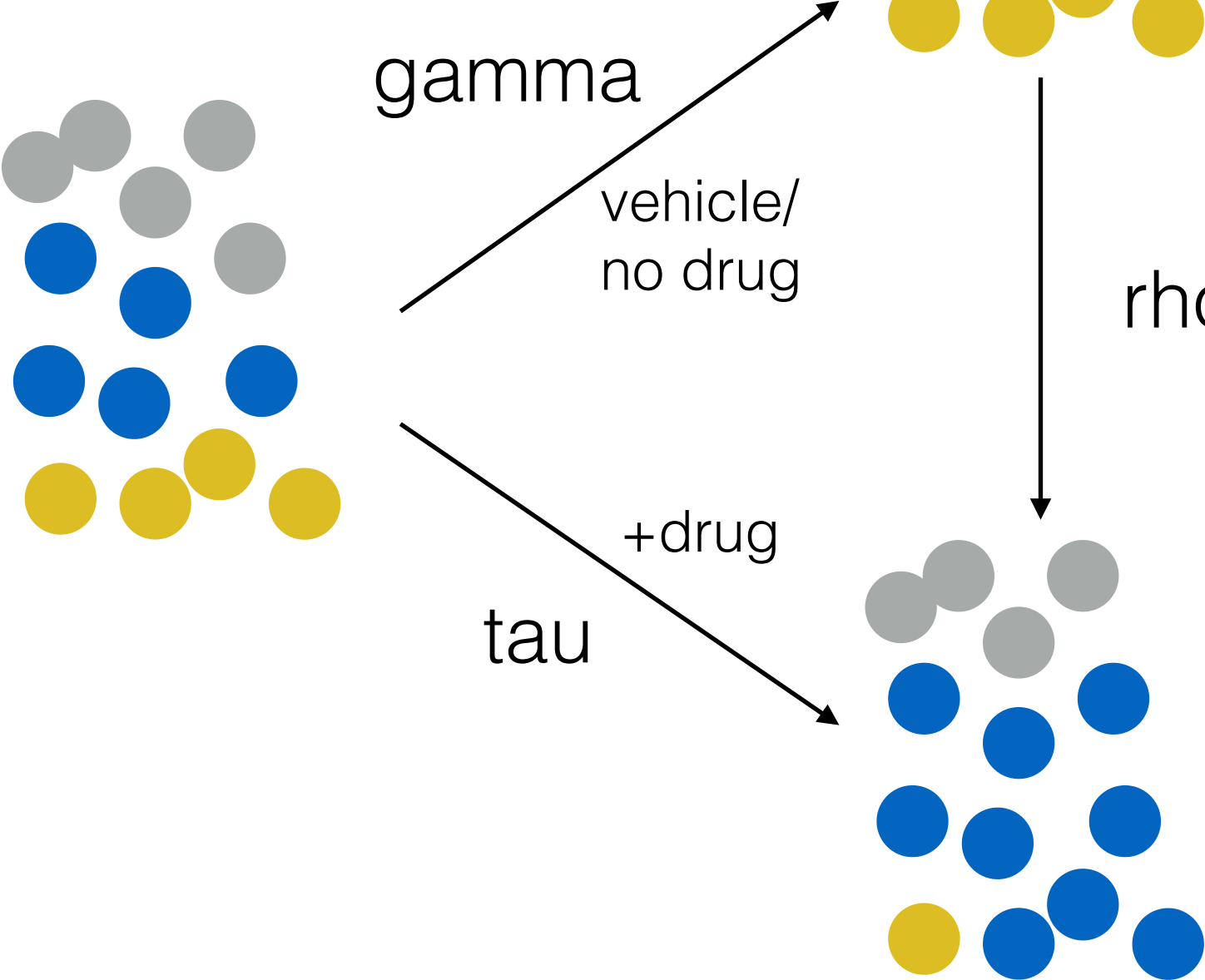
making custom graphs



We need to compare the sgRNA read counts from different conditions to obtain quantitative phenotypes

cells containing:
non-targeting sgRNA
sgRNA that promotes
resistance

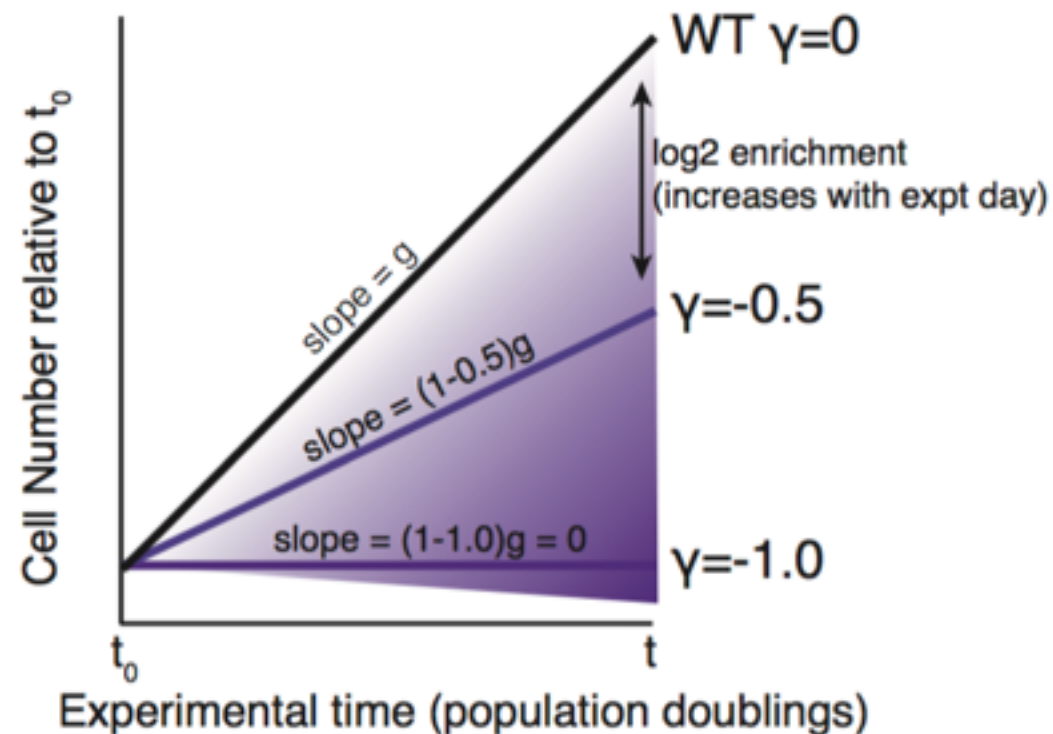
sensitivity



$\rho = \log_2\left(\frac{\text{fraction sgRNA counts, treated}}{\text{fraction sgRNA counts, untreated}}\right)$

Normalizing phenotype by cell doublings

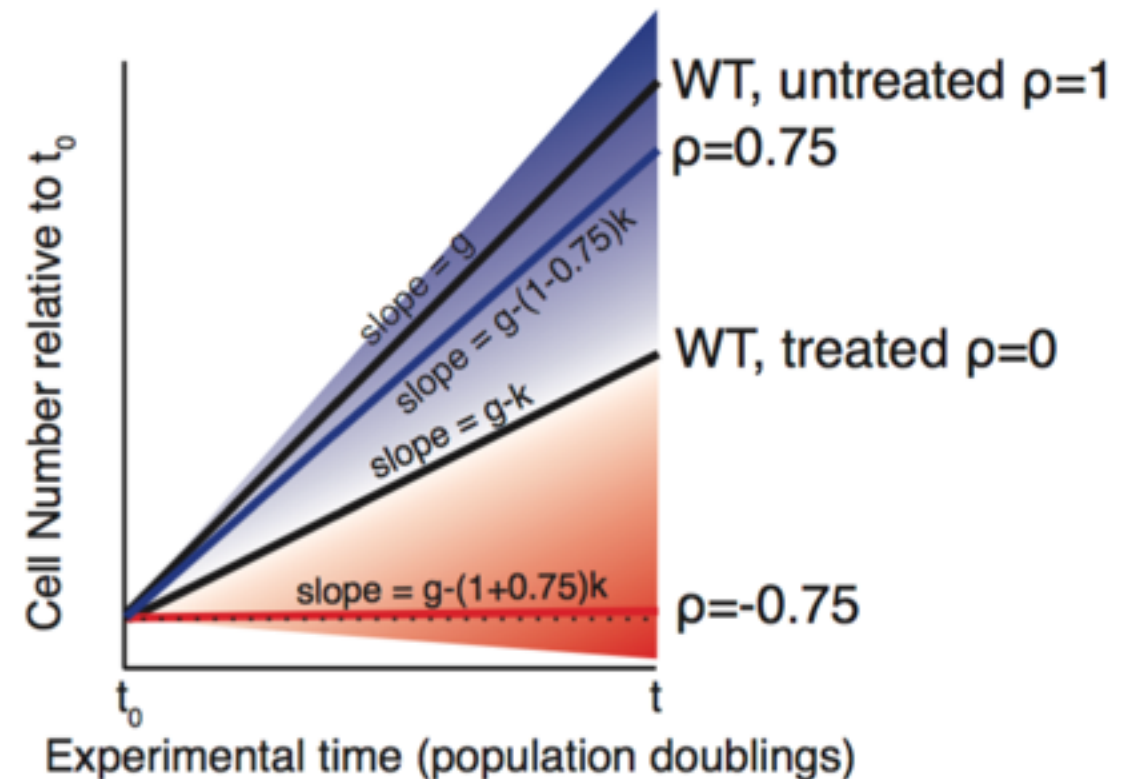
Growth Phenotypes



$\log_2 \text{enrichment} = \log_2(\text{Cell Fraction}_t / \text{Cell Fraction}_{t_0})$
 Normalized $\log_2 e = \text{Sample } \log_2 e - \text{Median NC } \log_2 e$

$\gamma = \text{Normalized } \log_2 e / t$

Toxin Phenotypes



$\log_2 \text{enrichment} = \log_2(\text{Cell Fraction}_{\text{treated}} / \text{Cell Fraction}_{\text{untreated}})$
 Normalized $\log_2 e = \text{Sample } \log_2 e - \text{Median NC } \log_2 e$

$\rho = \text{Normalized } \log_2 e / t$

See the templates in cell_doubling_measurements.xlsx to calculate t (number of cell doublings)

Setting up for process_experiments

▼ **Demo**

- ▶ **Step1**
- ▼ **Step2**
 - ▼ **count_files**
 - Demo_index1_full.counts
 - Demo_index3_full.counts
 - Demo_index6_full.counts
 - Demo_index10_full.counts
 - Demo_index12_full.counts
 - Demo_index14_full.counts
 - experiment_config_file_DEMO.txt
- ▶ **Step3**

Apoptosis+Cancer+Other_Cancer=A2M+_9268488.25-all~e39m1
174
Apoptosis+Cancer+Other_Cancer=A2M+_9268495.24-all~e39m1
84
Apoptosis+Cancer+Other_Cancer=A2M+_9268513.26-all~e39m1
168
Apoptosis+Cancer+Other_Cancer=A2M+_9268524.25-all~e39m1
181
Apoptosis+Cancer+Other_Cancer=A2M+_9268659.23-all~e39m1
83
Apoptosis+Cancer+Other_Cancer=A2M+_9268708.26-all~e39m1
18
Apoptosis+Cancer+Other_Cancer=A2M+_9268728.25-all~e39m1
109
Apoptosis+Cancer+Other_Cancer=A2M-_9268620.25-all~e39m1
198
Apoptosis+Cancer+Other_Cancer=A2M-_9268625.25-all~e39m1
191
Apoptosis+Cancer+Other_Cancer=A2M-_9268688.24-all~e39m1
106

Open the experiment_config_file
in a text editor

Setting up for process_experiments

```
#####  
##                Experiment Settings                ##  
#####  
[experiment_settings]
```

```
output_folder = [path to Step2-counts_to_gene_scores]  
experiment_name = ctx_demo
```

```
#####  
##                Experiment Settings                ##  
#####  
[experiment_settings]
```

```
output_folder = /home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output  
experiment_name = ctx_demo
```

Setting up for process_experiments

```
#####  
##                      Library Settings                      ##  
#####  
[library_settings]  
  
#currently CRISPRi_v1 or CRISPRa_v1  
|  
library = CRISPRi_v1  
  
#If you are using all sublibraries you don't need to change anything  
#Otherwise, uncomment and edit a list from below:  
  
#CRISPRi_v1 sublibraries  
sublibraries =  
    Apoptosis+Cancer+Other_Cancer  
#    Drug_Targets+Kinase_Phosphatase  
#    Gene_Expression  
#    Membrane_Proteins  
#    Stress_Proteostasis  
#    Trafficking+Mitochondria+Motility  
#    Unassigned  
#    Essential_CRISPRi  
  
#CRISPRa_v1 sublibraries  
#sublibraries =  
#    Apoptosis+Cancer+Other_Cancer  
#    Drug_Targets+Kinase_Phosphatase  
#    Gene_Expression  
#    Membrane_Proteins  
#    Stress_Proteostasis  
#    Trafficking+Mitochondria+Motility  
#    Unassigned
```

= ignore this line, so only the
Apoptosis+Cancer+Other_Cancer sub-library
will be analyzed to make this demo faster

Setting up for process_experiments

```
#####
##                               Counts Files                               ##
#####
[counts_files]

#Enter the paths to counts files from fastqgz_to_counts.py
#followed by :condition|replicate_id
#For example:
#counts_file_string =
#   /home/max/counts/Sample1_index6_CRISPRi.counts:treated|Rep1
#   /home/max/counts/Sample1_index12_CRISPRi.counts:untreated|Rep1

counts_file_string =
[index 10]:T0|Rep1
[index 3]:untreated|Rep1
[index 1]:treated|Rep1
[index 12]:T0|Rep2
[index 6]:untreated|Rep2
[index 14]:treated|Rep2

counts_file_string =
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/count_files/Demo_index10_full.counts:T0|Rep1
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/count_files/Demo_index3_full.counts:untreated|Rep1
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/count_files/Demo_index1_full.counts:treated|Rep1
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/count_files/Demo_index12_full.counts:T0|Rep2
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/count_files/Demo_index6_full.counts:untreated|Rep2
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/count_files/Demo_index14_full.counts:treated|Rep2
```

Once you've made these changes, save the file (or save as
experiment_config_filledout.txt)

Setting up for process_experiments

```
#####  
##          Growth Values (phenotype scores only)      ##  
#####  
[growth_values]  
#Enter the growth values (population doublings/doubling differences)  
#These values are used to normalize log2enrichments;  
#default to 1 (un-normalized) if log2e is desired  
  
#comparison_name:replicate_id:value  
#For example:  
#growth_value_string =  
#  rho:Rep1:7.13  
  
growth_value_string =  
  gamma:Rep2:11.1519761622  
  rho:Rep2:8.44158496881  
  tau:Rep2:2.7103911934  
  gamma:Rep1:10.7412001484  
  rho:Rep1:7.82935376641  
  tau:Rep1:2.9118463821
```

Note that this demo has pre-filled values for the growth_value_string

- This allows you to normalize phenotypes by cell doublings (see previous slides)
- Growth values can be calculated using the cell doubling measurements spreadsheet
- This normalization is optional for growth screens and may not be applicable for other types of screens

Once you've made these changes, save the file (or save as
experiment_config_filledout.txt)

Running process_experiments

```
[In [8]: run process_experiments.py -h
```

```
usage: process_experiments.py [-h] Config_File Library_File_Directory
```

```
Calculate sgRNA- and gene-level phenotypes based on sequencing read counts, as  
specified by the experiment config file.
```

```
positional arguments:
```

```
  Config_File          Experiment config file specifying screen analysis  
                        settings (see accomapnying BLANK and DEMO files).
```

```
  Library_File_Directory Directory containing reference library tables and the  
                        library_config.txt file.
```

```
optional arguments:
```

```
  -h, --help          show this help message and exit
```

Running process_experiments

```
[In [8]: run process_experiments.py -h
```

```
usage: process_experiments.py [-h] Config_File Library_File_Directory
```

```
Calculate sgRNA- and gene-level phenotypes based on sequencing read counts, as  
specified by the experiment config file.
```

```
positional arguments:
```

```
  Config_File          Experiment config file specifying screen analysis  
                        settings (see accomapnying BLANK and DEMO files).
```

```
  Library_File_Directory Directory containing reference library tables and the  
                        library_config.txt file.
```

```
optional arguments:
```

```
  -h, --help          show this help message and exit
```

```
[In [9]: run process_experiments.py Demo/Step2/experiment_config_file_DEMO_filledout.txt library_tables/
```

Running process_experiments

```
[In [8]: run process_experiments.py -h
```

```
usage: process_experiments.py [-h] Config_File Library_File_Directory
```

```
Calculate sgRNA- and gene-level phenotypes based on sequencing read counts, as  
specified by the experiment config file.
```

```
positional arguments:
```

```
  Config_File          Experiment config file specifying screen analysis  
                        settings (see accomapnying BLANK and DEMO files).
```

```
  Library_File_Directory Directory containing reference library tables and the  
                        library_config.txt file.
```

```
optional arguments:
```

```
  -h, --help          show this help message and exit
```

```
[In [9]: run process_experiments.py Demo/Step2/experiment_config_file_DEMO_filledout.txt library_tables/
```

```
Accessing library information
```

```
Loading counts data
```

```
Merging experiment counts split across lanes/indexes
```

Running process_experiments

```
[In [8]: run process_experiments.py -h
```

```
usage: process_experiments.py [-h] Config_File Library_File_Directory
```

```
Calculate sgRNA- and gene-level phenotypes based on sequencing read counts, as  
specified by the experiment config file.
```

```
positional arguments:
```

```
  Config_File          Experiment config file specifying screen analysis  
                        settings (see accomapnying BLANK and DEMO files).
```

```
  Library_File_Directory Directory containing reference library tables and the  
                        library_config.txt file.
```

```
optional arguments:
```

```
  -h, --help          show this help message and exit
```

```
[In [9]: run process_experiments.py Demo/Step2/experiment_config_file_DEMO_filledout.txt library_tables/
```

```
Accessing library information
```

```
Loading counts data
```

```
Merging experiment counts split across lanes/indexes
```

Once you see that the script is running,
it will take ~5 minutes to complete...

Running process_experiments

```
[In [9]: run process_experiments.py Demo/Step2/experiment_config_file_DEMO_filledout.txt library_tables/
```

```
Accessing library information
Loading counts data
Merging experiment counts split across lanes/indexes
-generating sgRNA read count histograms
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/000_fig_counts_hist.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/001_fig_counts_hist.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/002_fig_counts_hist.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/003_fig_counts_hist.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/004_fig_counts_hist.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/005_fig_counts_hist.svg
Computing sgRNA phenotype scores
-generating phenotype histograms and scatter plots
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/006_fig_counts_scatter.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/007_fig_phenotype_hist.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/008_fig_sgRNAs_passing_filter_hist.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/009_fig_counts_scatter.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/010_fig_phenotype_hist.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/011_fig_sgRNAs_passing_filter_hist.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/012_fig_counts_scatter.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/013_fig_phenotype_hist.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/014_fig_sgRNAs_passing_filter_hist.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/015_fig_counts_scatter.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/016_fig_phenotype_hist.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/017_fig_sgRNAs_passing_filter_hist.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/018_fig_counts_scatter.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/019_fig_phenotype_hist.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/020_fig_sgRNAs_passing_filter_hist.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/021_fig_counts_scatter.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/022_fig_phenotype_hist.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/023_fig_sgRNAs_passing_filter_hist.svg
Averaging replicates
-generating replicate phenotype histograms and scatter plots
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/024_fig_phenotype_scatter.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/025_fig_phenotype_hist.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/026_fig_phenotype_scatter.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/027_fig_phenotype_hist.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/028_fig_phenotype_scatter.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/029_fig_phenotype_hist.svg
Generating a pseudogene distribution from negative controls
Computing gene scores
--calculate_ave
--calculate_mw
Collapsing transcript scores to gene scores
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/030_fig_volcano_plot.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/031_fig_volcano_plot.svg
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/032_fig_volcano_plot.svg
Done!
```

Running process_experiments

```
[In [9]: run process_experiments.py Demo/Step2/experiment_config_file_DEMO_filledout.txt library_tables/
```

Accessing library information

Loading counts data

Merging experiment counts split across lanes/indexes

-generating sgRNA read count histograms

/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/000_fig_counts_hist.svg

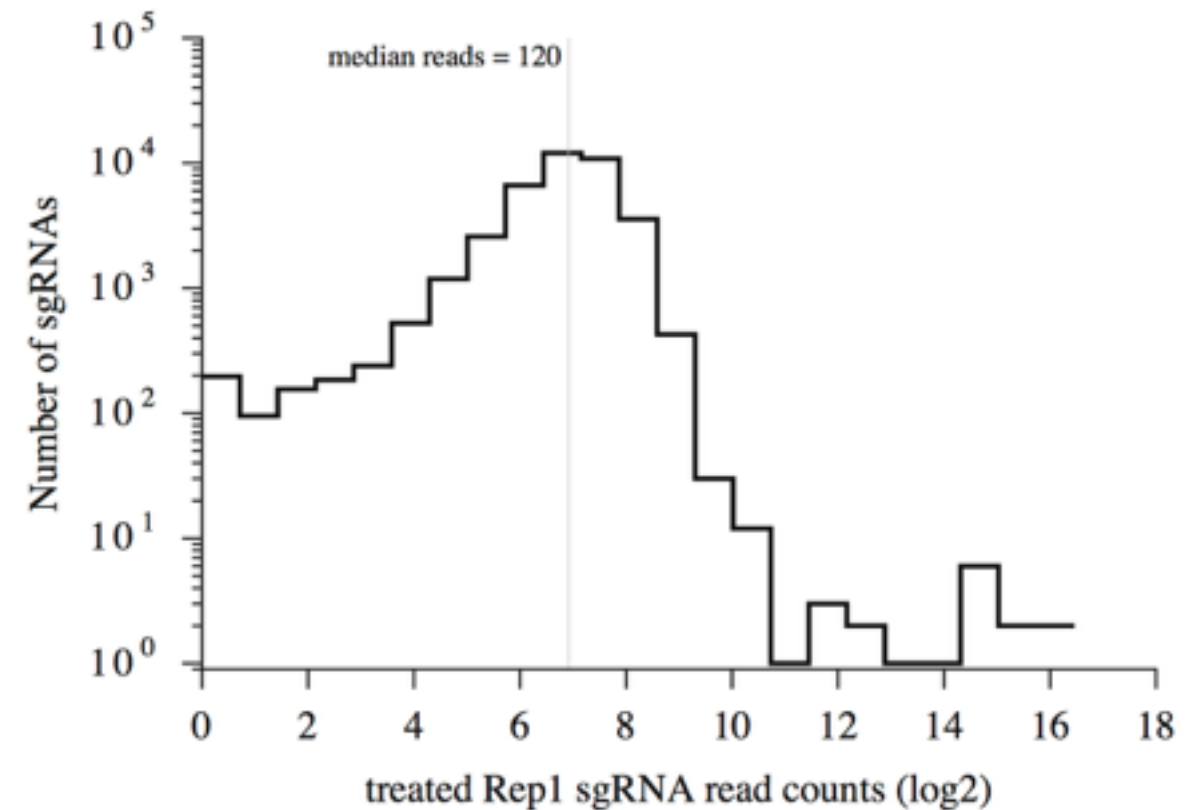
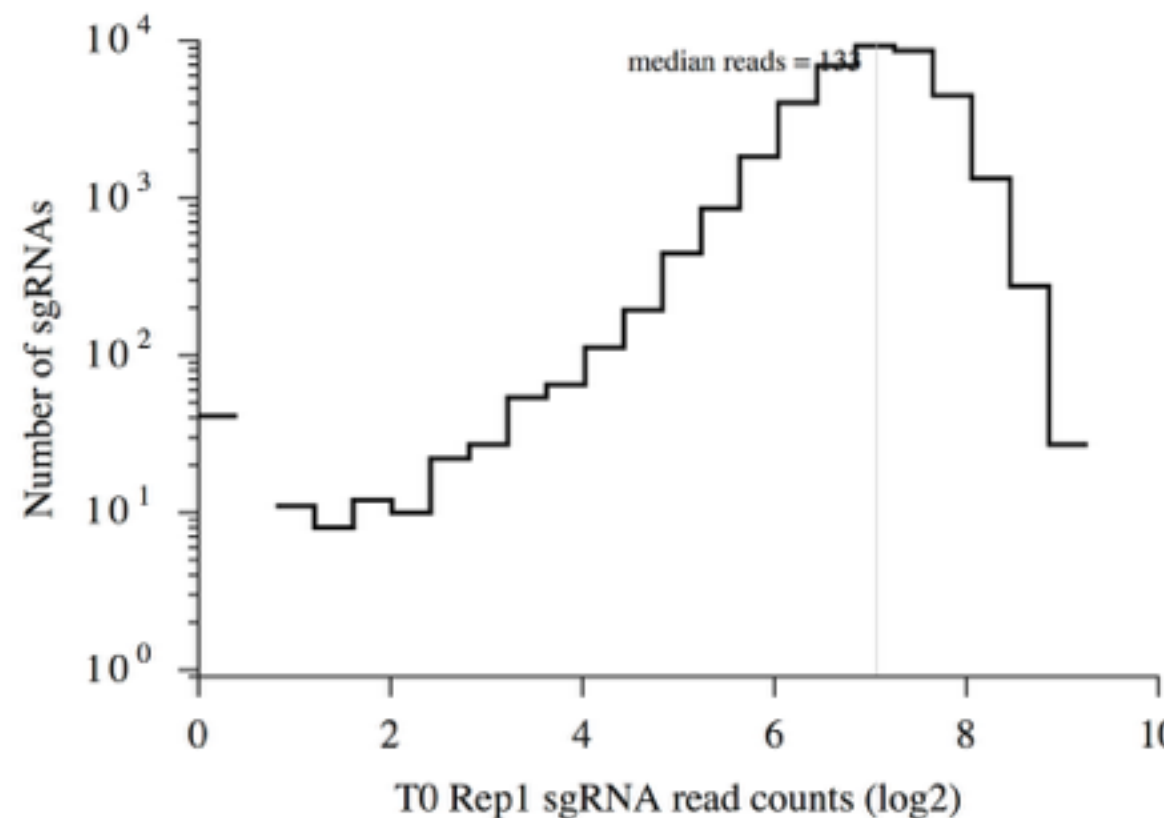
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/001_fig_counts_hist.svg

/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/002_fig_counts_hist.svg

/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/003_fig_counts_hist.svg

/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/004_fig_counts_hist.svg

/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/005_fig_counts_hist.svg

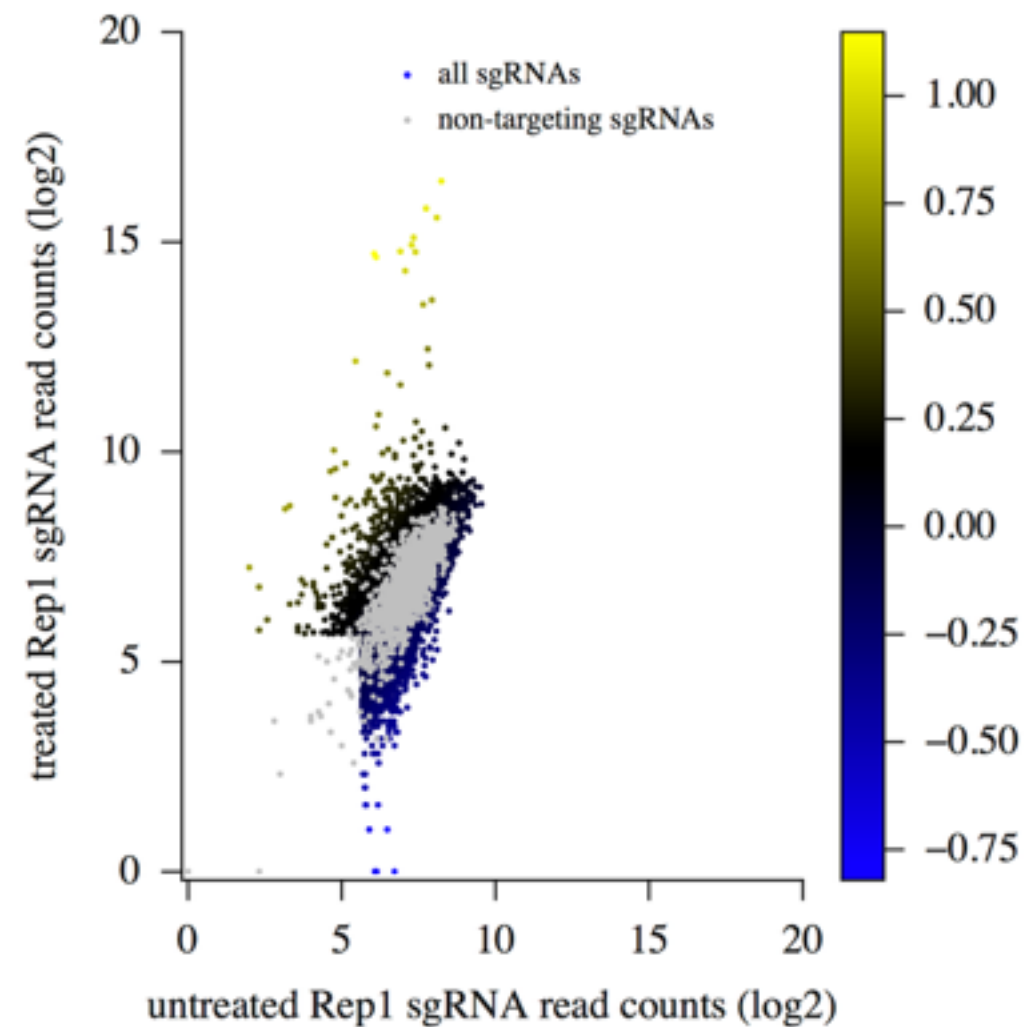
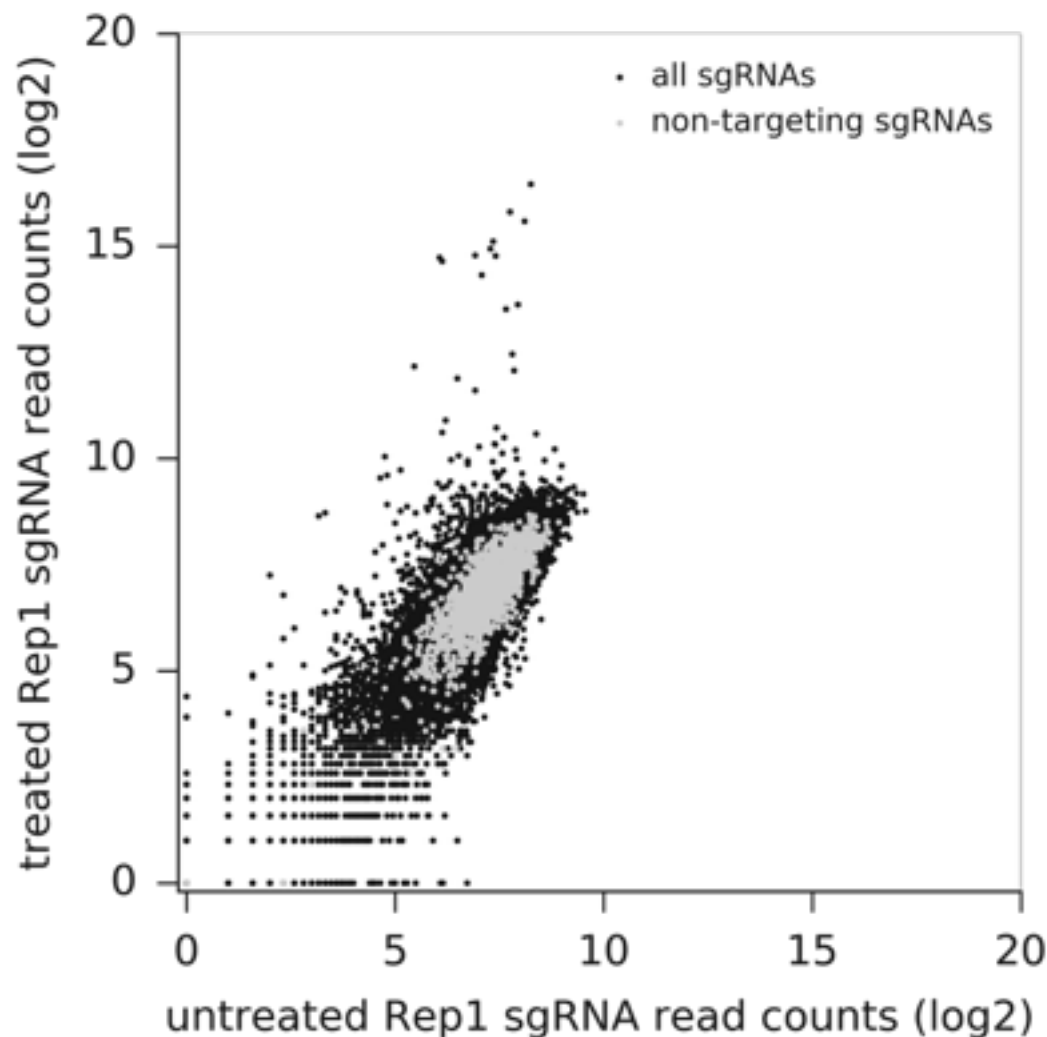


Running process_experiments

Computing sgRNA phenotype scores

-generating phenotype histograms and scatter plots

```
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/006_fig_counts_scatter.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/007_fig_phenotype_hist.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/008_fig_sgRNAs_passing_filter_hist.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/009_fig_counts_scatter.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/010_fig_phenotype_hist.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/011_fig_sgRNAs_passing_filter_hist.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/012_fig_counts_scatter.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/013_fig_phenotype_hist.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/014_fig_sgRNAs_passing_filter_hist.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/015_fig_counts_scatter.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/016_fig_phenotype_hist.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/017_fig_sgRNAs_passing_filter_hist.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/018_fig_counts_scatter.svg
```

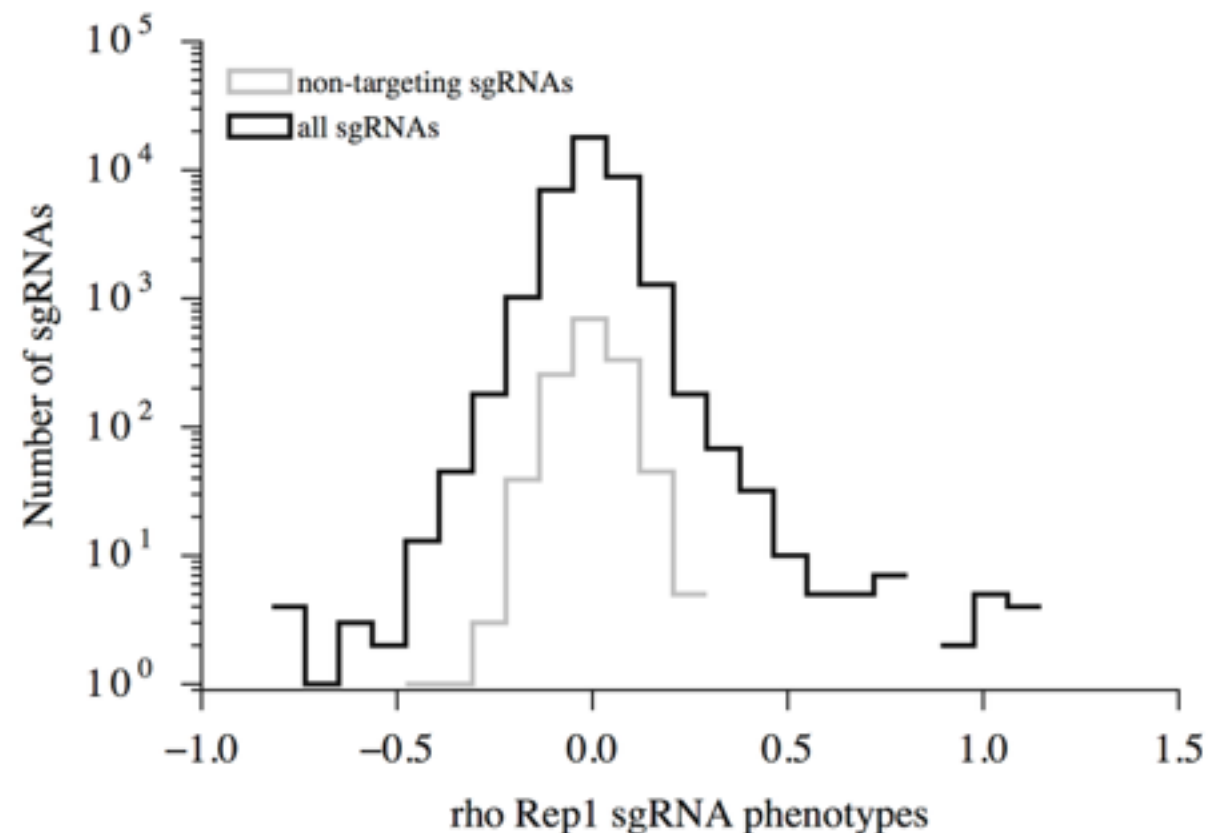


Running process_experiments

Computing sgRNA phenotype scores

-generating phenotype histograms and scatter plots

```
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/006_fig_counts_scatter.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/007_fig_phenotype_hist.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/008_fig_sgRNAs_passing_filter_hist.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/009_fig_counts_scatter.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/010_fig_phenotype_hist.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/011_fig_sgRNAs_passing_filter_hist.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/012_fig_counts_scatter.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/013_fig_phenotype_hist.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/014_fig_sgRNAs_passing_filter_hist.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/015_fig_counts_scatter.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/016_fig_phenotype_hist.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/017_fig_sgRNAs_passing_filter_hist.svg  
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/018_fig_counts_scatter.svg
```



Running process_experiments

Averaging replicates

-generating replicate phenotype histograms and scatter plots

`/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/024_fig_phenotype_scatter.svg`

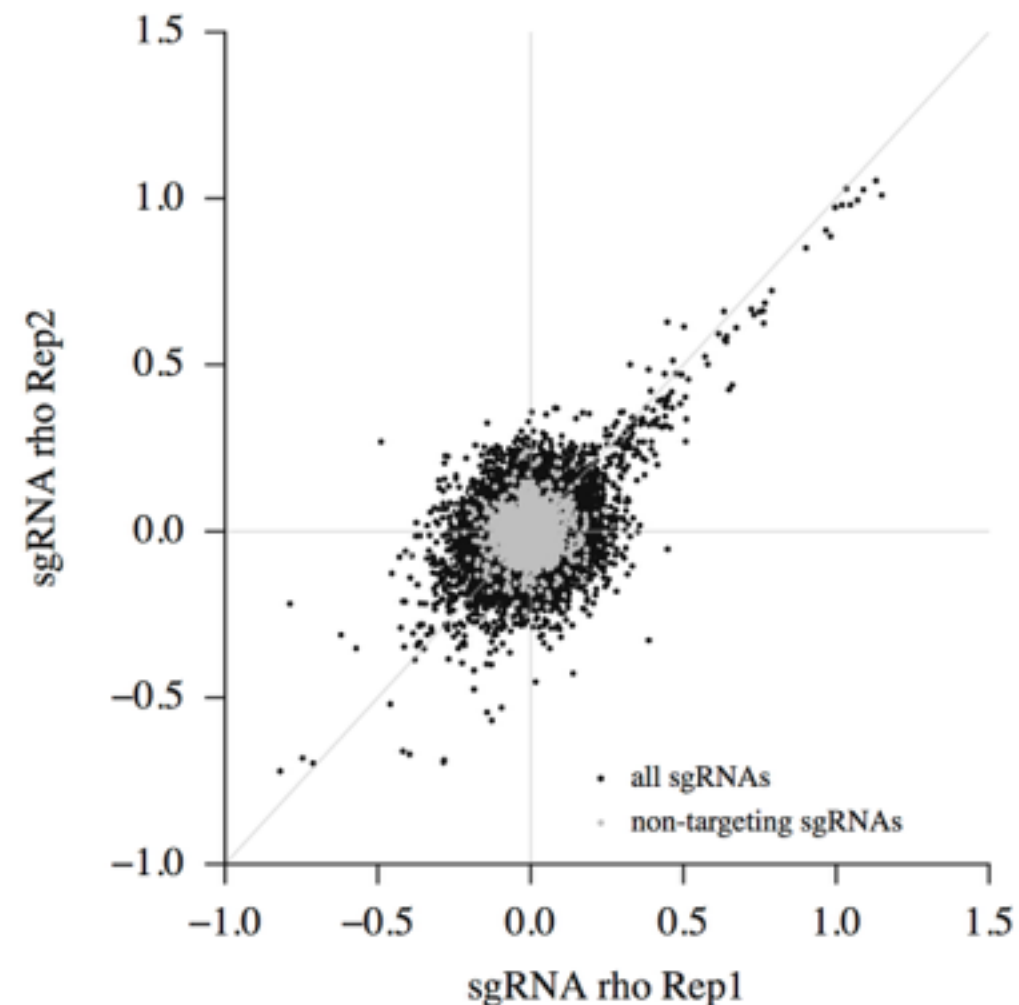
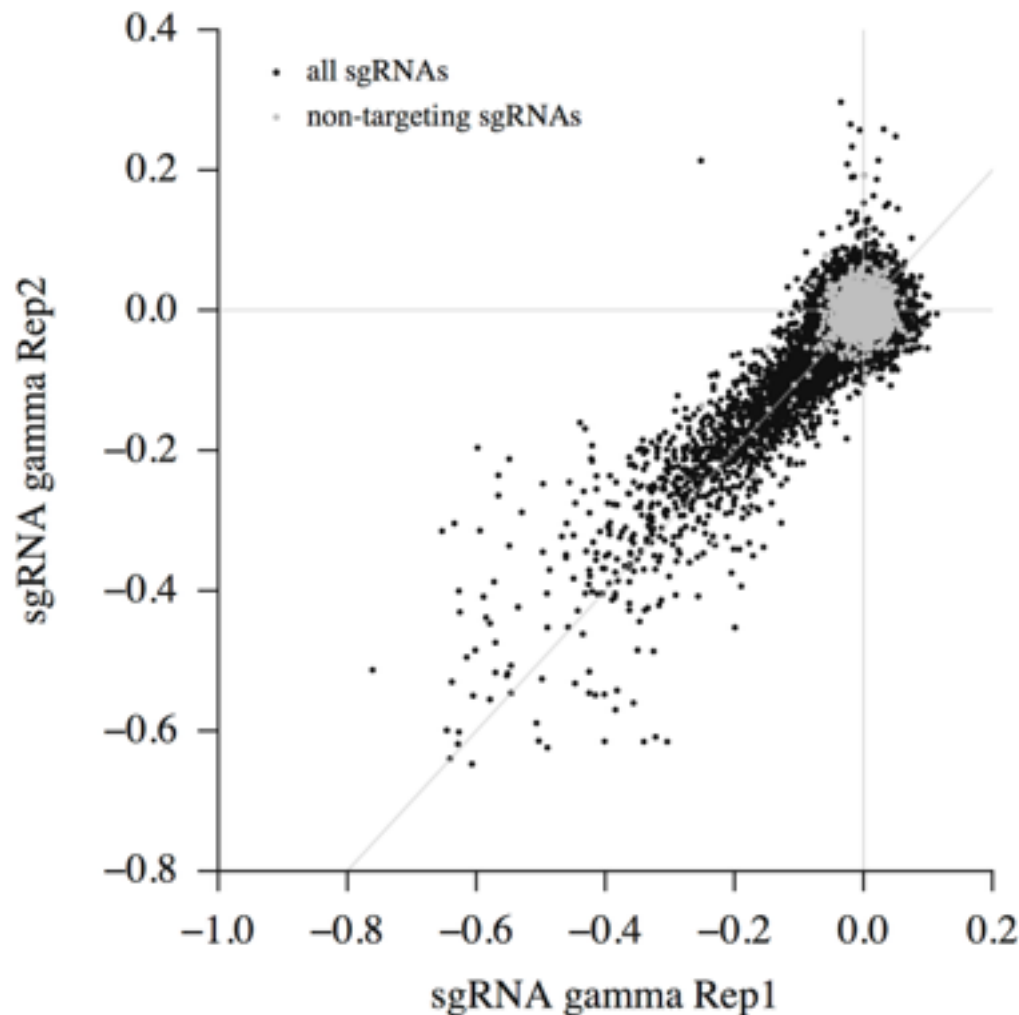
`/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/025_fig_phenotype_hist.svg`

`/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/026_fig_phenotype_scatter.svg`

`/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/027_fig_phenotype_hist.svg`

`/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/028_fig_phenotype_scatter.svg`

`/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/029_fig_phenotype_hist.svg`



Running process_experiments

Generating a pseudogene distribution from negative controls

Computing gene scores

```
--calculate_ave
```

```
--calculate_mw
```

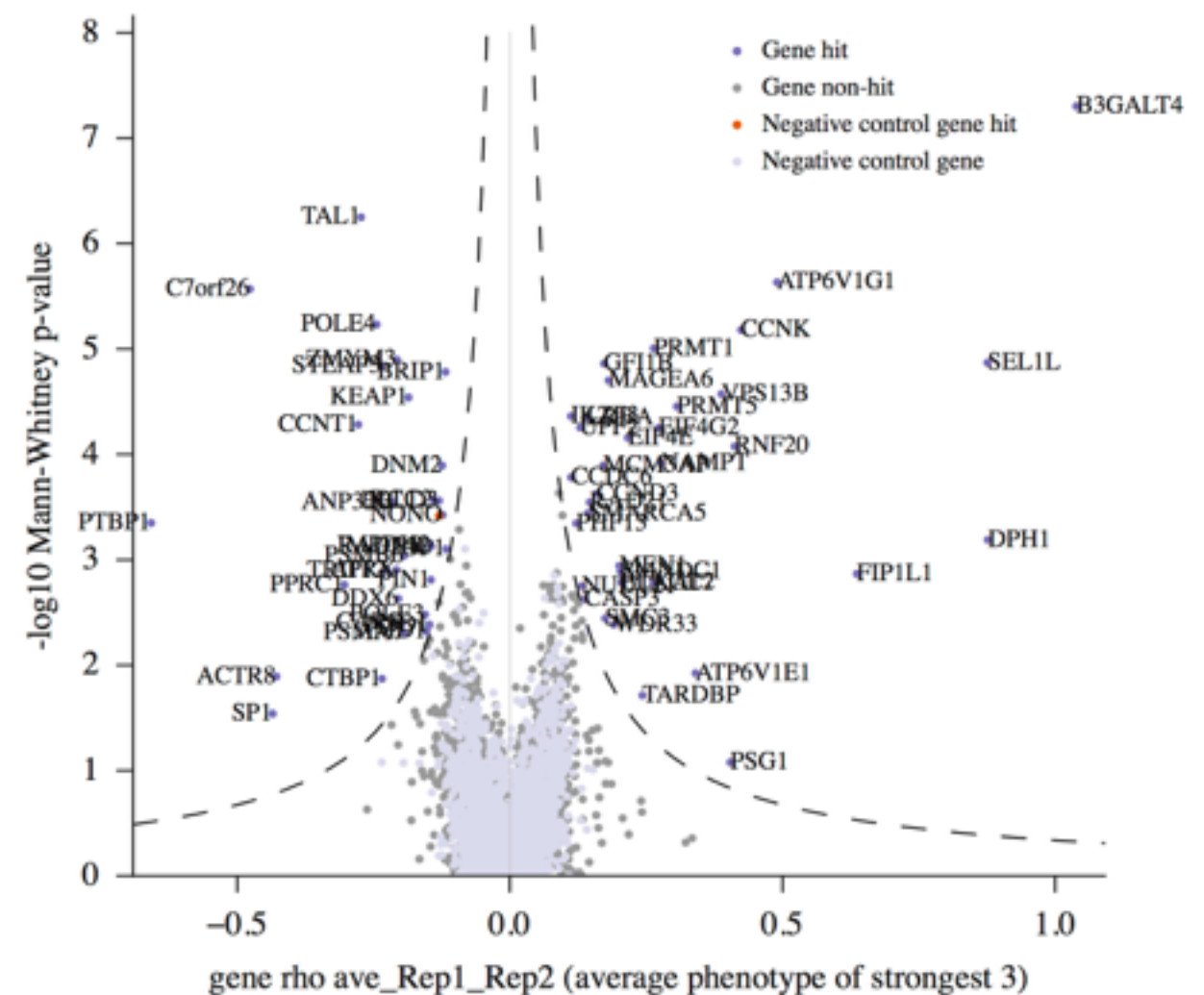
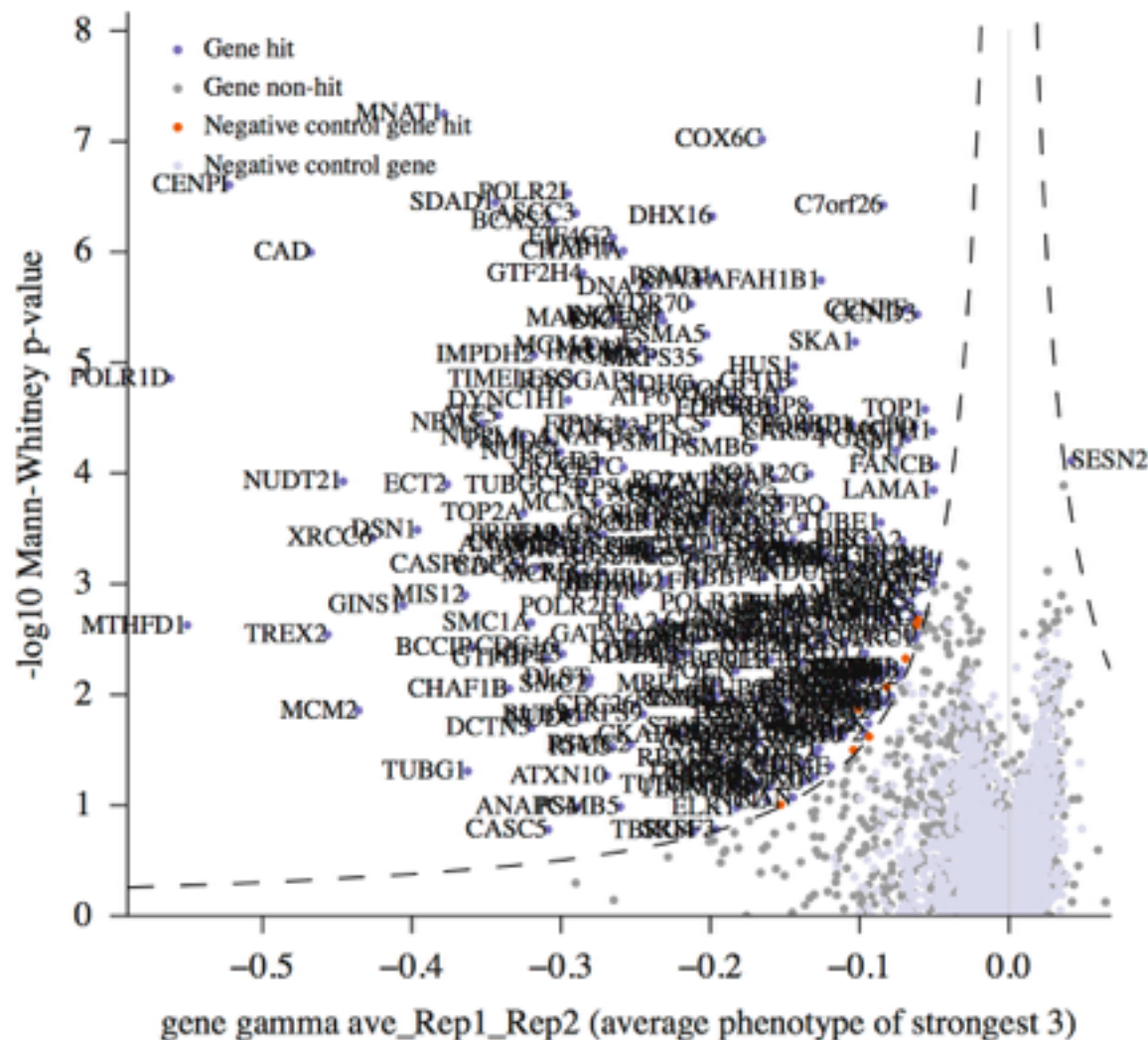
Collapsing transcript scores to gene scores

/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/030_fig_volcano_plot.svg

```
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/031_fig_volcano_plot.svg
```

```
/home/mhorlbeck/home_tmp/mhorlbeck/ScreenProcessing/Demo/Step2/output/ctx_demo_plots/032_fig_volcano_plot.svg
```

Done!



Step 1: fastqgz_to_counts.py

counting sgRNAs in raw sequencing files



Step 2: process_experiments.py

calculating sgRNA-level and gene-level phenotypes and p-values



Step 3: screen_analysis.py

making custom graphs

process_experiments resulted in a
lot of data tables....

▼ **Demo**

▶ **Step1**

▶ **Step2**

▼ **Step3**

 ctx_demo_full_genetable_collapsed.txt

 ctx_demo_full_genetable.txt

 ctx_demo_full_librarytable.txt

 ctx_demo_full_mergedcountstable_summary.txt

 ctx_demo_full_mergedcountstable.txt

 ctx_demo_full_phenotypetable.txt

 ctx_demo_full_rawcountstable_summary.txt

 ctx_demo_full_rawcountstable.txt

...which can be explored using excel or similar programs

▼ **Demo**

▶ **Step1**

▶ **Step2**

▼ **Step3**

ctx_demo_full_genetable_collapsed.txt

ctx_demo_full_genetable.txt

ctx_demo_full_librarytable.txt

ctx_demo_full_mergedcountstable_summary.txt

ctx_demo_full_mergedcountstable.txt

ctx_demo_full_phenotypetable.txt

ctx_demo_full_rawcountstable_summary.txt

ctx_demo_full_rawcountstable.txt

final gene-level results

information on the sgRNAs
present in the library

sgRNA-level phenotype scores
calculated as $\log_2$ enrichment
(normalized by growth values if
provided)

raw read counts of sgRNAs before and after
merging multiple counts files assigned to the
same sample condition/replicate

Interpreting genetable_collapsed

	A	B	C	D	E	F	
1		gamma	gamma	gamma	gamma	gamma	g
2		Rep1	Rep1	Rep1	Rep1	Rep1	F
3		transcripts	Mann-Whitney p-value	average phenotype of strongest 3	sgRNA count_MW	sgRNA count_avg	t
4	gene						
5	A1BG	all	0.617489319	0.008079268	10	10	a
6	A1CF	ENST00000374001.2,EI	0.774850008	-0.002736356	9	9	E
7	A2M	all	0.072458779	0.013644911	10	10	a
8	A2ML1	all	0.739439742	-0.01119083	9	9	a
9	A4GALT	ENST00000249005.2,EI	0.499943531	-0.00265974	10	10	E

Transcription start site with lowest-scoring p-value, where multiple exist (all transcripts are listed in _genetable.txt)

The phenotypes of all sgRNAs targeting the gene are ranked by absolute value, and the top 3 are averaged to obtain a gene-level phenotype

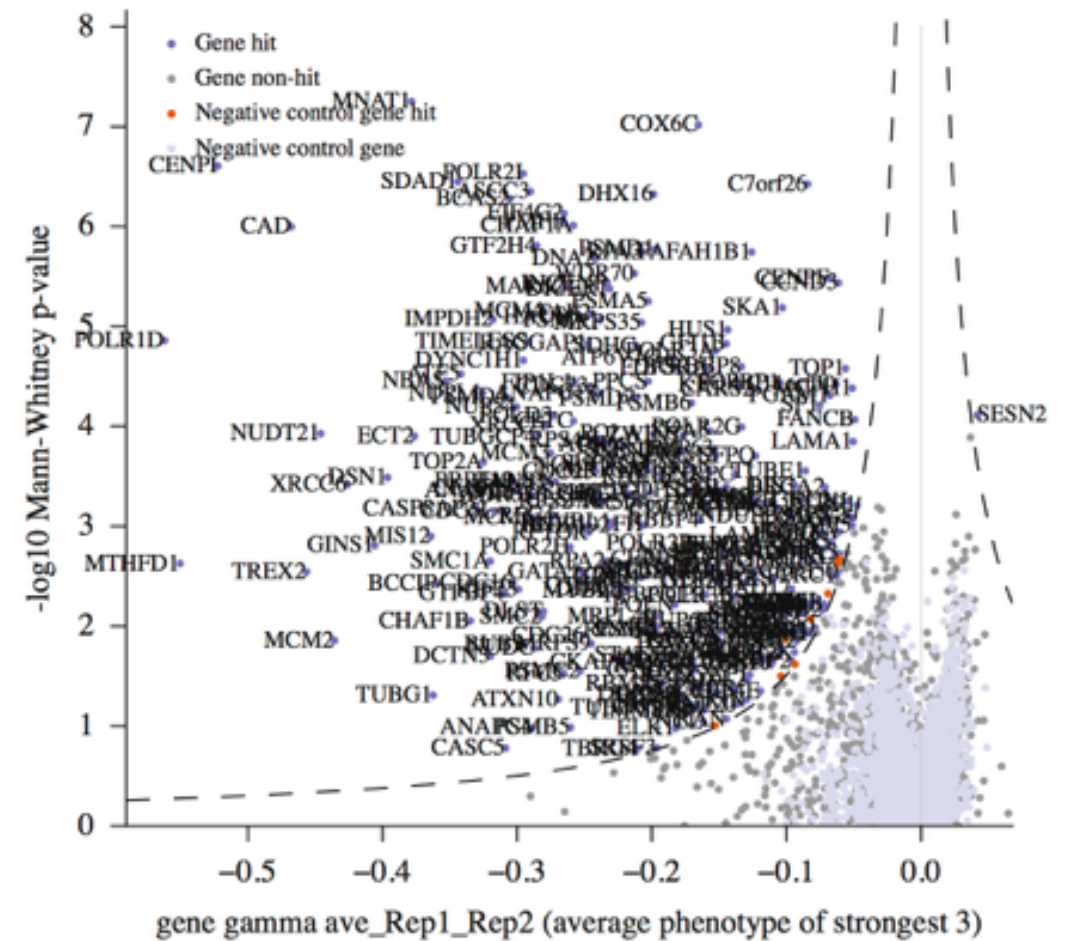
Mann-Whitney p-value comparing the phenotypes of all sgRNAs targeting the gene to all negative control sgRNAs

Number of sgRNAs included in the library and passing the read count filter

Interpreting genetable_collapsed

	A	B	C	D	E	F	G
1		gamma	gamma	gamma	gamma	gamma	g
2		Rep1	Rep1	Rep1	Rep1	Rep1	F
3		transcripts	Mann-Whitney p-value	average phenotype of strongest 3	sgRNA count_MW	sgRNA count_avg	t
4	gene						
5	A1BG	all	0.617489319	0.008079268	10	10	a
6	A1CF	ENST00000374001.2,El	0.774850008	-0.002736356	9	9	E
7	A2M	all	0.072458779	0.013644911	10	10	a
8	A2ML1	all	0.739439742	-0.01119083	9	9	a
9	A4GALT	ENST00000249005.2,El	0.499943531	-0.00265974	10	10	E

The p-value and phenotype scores provide complementary information to guide follow-up of screen hits



screen_analysis offers a number of functions to generate new graphs

```
[In [10]: run screen_analysis.py
```

```
[In [11]: data = loadData('Demo/Step3/ctx_demo_full')
```

all functions follow the format:
graphName(data, phenotype, replicate, ...)

and you can enter graphName(data) to find out what the available phenotype/replicate options are for the plot

```
[In [12]: countsHistogram(data)
Condition and Replicate options are:
T0          Rep1
T0          Rep2
treated     Rep1
treated     Rep2
untreated   Rep1
untreated   Rep2
```

Available plotting functions

loadData(**experimentName**, collapsedToTranscripts = True, premergedCounts = False)

countsHistogram(data, **condition**=None, **replicate**=None)

countsScatter(data, **condition_x** = None, **replicate_x** = None,
 condition_y = None, **replicate_y** = None,
 showAll = True, showNegatives = True, showGenes = [],
 colorByPhenotype_condition = None, colorByPhenotype_replicate = None)

premergedCountsScatterMatrix(data, condition=None, replicate=None)

phenotypeHistogram(data, **phenotype**=None, **replicate**=None)

phenotypeScatter(data, **phenotype_x** = None, **replicate_x** = None,
 phenotype_y = None, **replicate_y** = None,
 showAll = True, showNegatives = True,
 showGenes = [], showGeneSets = {})

sgRNAsPassingFilterHist(data, phenotype, replicate, transcripts=False)

volcanoPlot(data, **phenotype**=None, **replicate**=None, transcripts=False, showPseudo=True,
 effectSizeLabel=None, pvalueLabel=None, hitThreshold=7,
 labelHits = False, showGeneSets = {}, labelGeneSets = True)